

Morphological Image Processing for Binary Images

Morphological Image Processing

1. It is a tool for extracting image components that are useful in the representation and description of region shape, such as boundaries, skeletons, etc.
2. We use the language in the Set theory to represent morphological operations.
3. Link for more details:

http://en.wikipedia.org/wiki/Mathematical_morphology

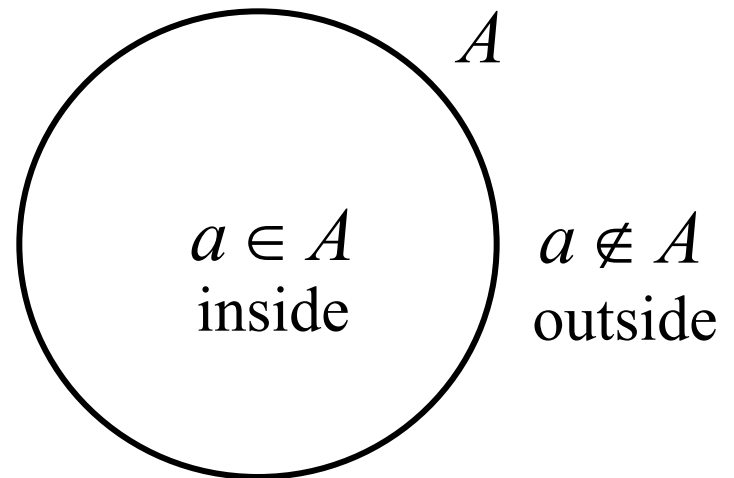
Basic Concepts from Set Theory

1. Let A be a set in Z^2 (2D integer space)
2. If $a = (a_1, a_2)$ is an element of A , where a_1 and a_2 are coordinates of pixels in 2D, then we write

$$a \in A$$

3. If a is not an element of A , then we write

$$a \notin A$$



Basic Concepts from Set Theory

4. If A is a set with no element, then we call it *null or empty set*.

$$A = \emptyset$$

5. A set is specified by the contents of two braces: $\{ \}$,

a. e.g.,

$$C = \{ w \mid w = -d, \text{ for } d \in D \}$$

- b. C represents a set of elements.
c. w represents the elements.
d. All elements inside C are formed by multiplying the element coordinates, d , by -1 .

Basic Concepts from Set Theory

6. A is a subset of B : every element of set A is also an element of set B . (Every element of A is in B)

$$A \subseteq B$$

7. Union of two sets A and B : the set of all elements belonging to either A or B . (All elements are either in A or B .)

$$C = A \cup B$$

Basic Concepts from Set Theory

8. Intersection of two sets A and B : the set of all elements belonging to both A and B . (All elements are in A and B .)

$$D = A \cap B$$

9. Two sets A and B are disjoint or mutually exclusive if their intersection is an empty set.

$$A \cap B = \emptyset$$

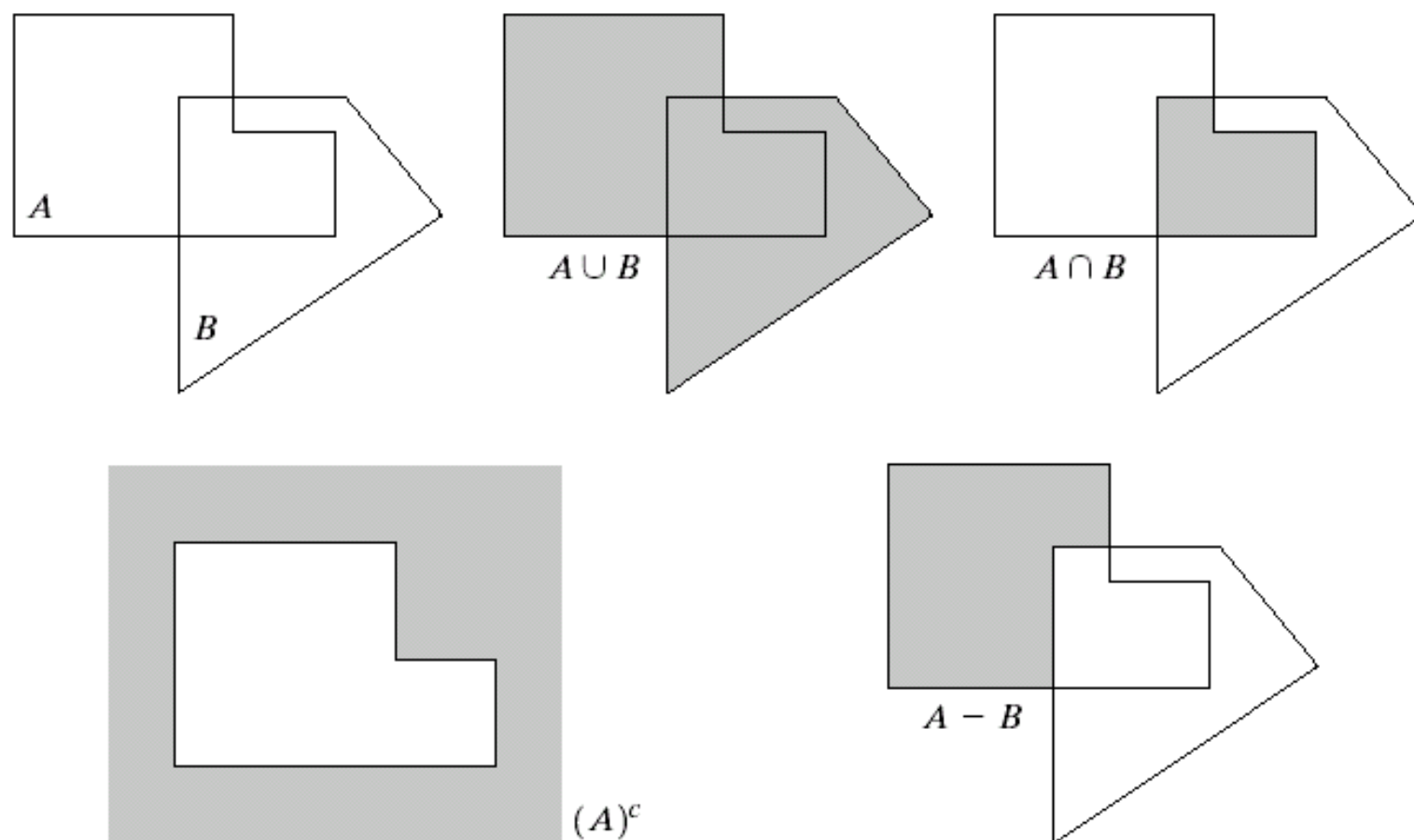
Basic Concepts from Set Theory

10. Complement of a set A : the set of elements not contained in A .

$$A^c = \{w \mid w \notin A\}$$

11. Difference of two sets A and B : the set of all elements belonging to A and not contained in B .

$$A - B = \{w \mid w \in A, w \notin B\} = A \cap B^c$$



a	b	c
d	e	

FIGURE 9.1

(a) Two sets A and B . (b) The union of A and B . (c) The intersection of A and B . (d) The complement of A . (e) The difference between A and B .

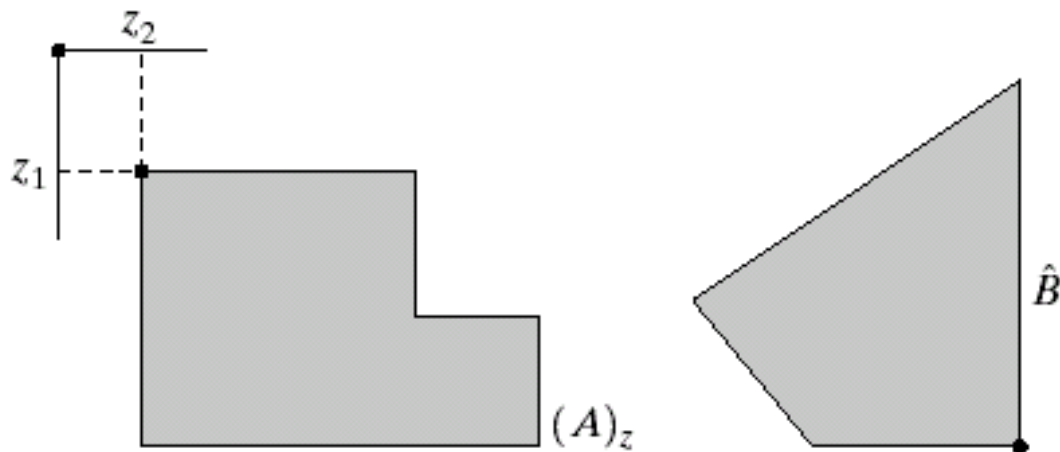
Basic Concepts from Set Theory

12. Reflection of set B:

$$\hat{B} = \{w \mid w = -b, \text{ for } b \in B\}$$

13. Translation of set A:

$$(A)_z = \{c \mid c = a + z, \text{ for } a \in A\}$$



a b

FIGURE 9.2

(a) Translation of A by z .

(b) Reflection of B . The sets A and B are from Fig. 9.1.

Digital Representation of the Images and Sets

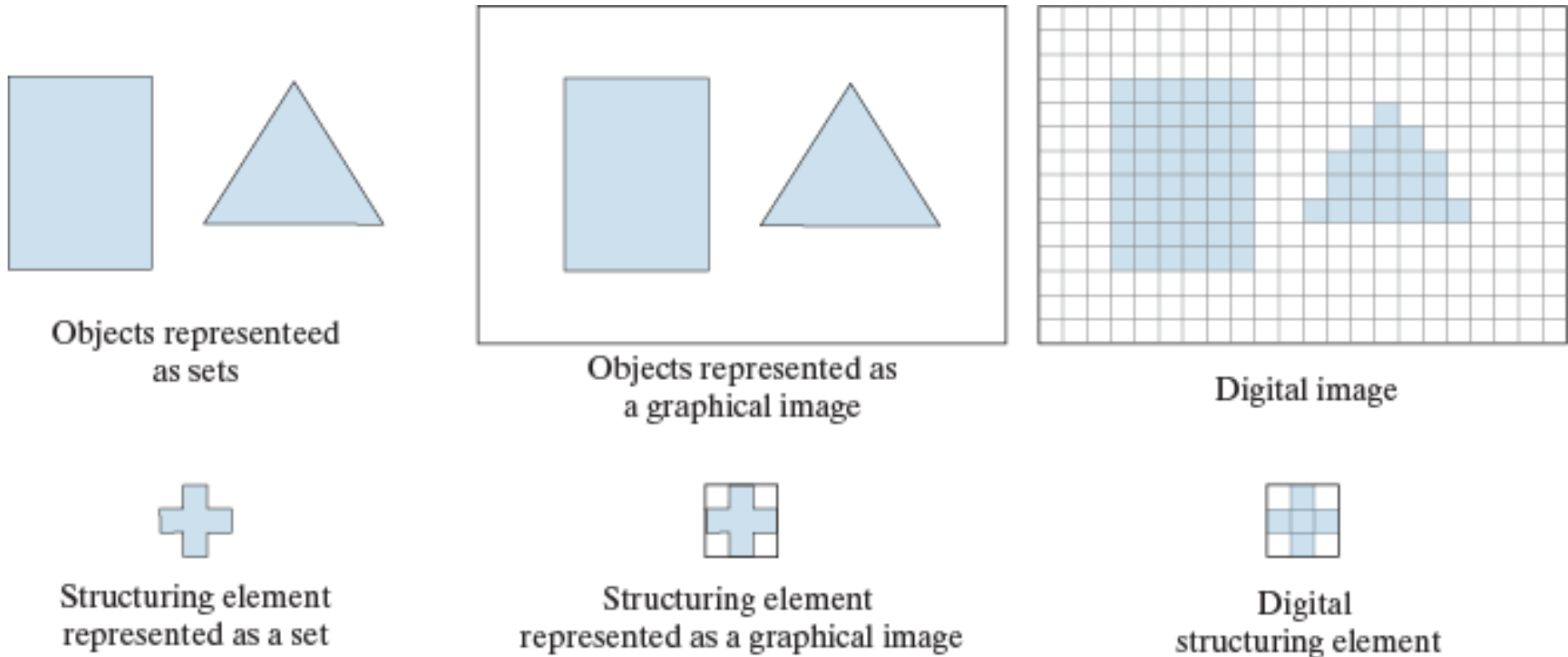
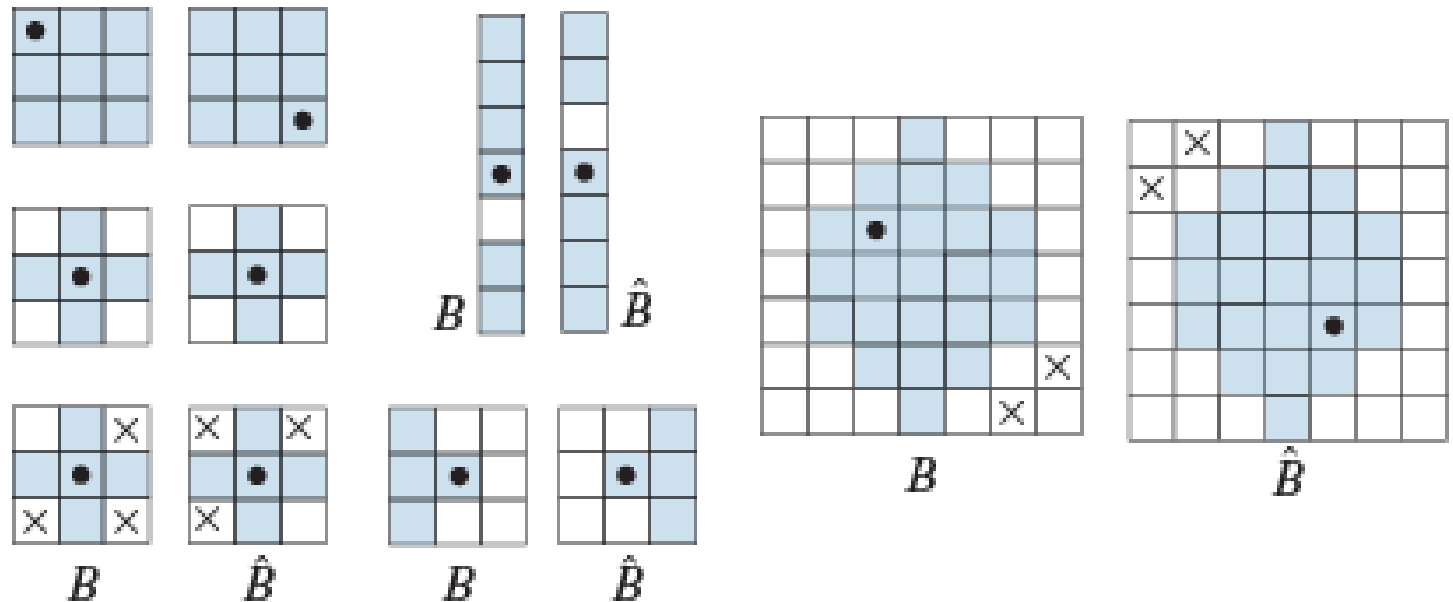


FIGURE 9.1 Top row. *Left:* Objects represented as graphical sets. *Center:* Objects embedded in a background to form a graphical image. *Right:* Object and background are digitized to form a digital image (note the grid). Second row: Example of a structuring element represented as a set, a graphical image, and finally as a digital SE.

Reflections of the Structuring Elements

FIGURE 9.2
Structuring elements and their reflections about the origin (the $\times$'s are don't care elements, and the dots denote the origin). Reflection is rotation by 180° of an SE about its origin.



Logic Operations Involving Binary Images

1. All images are binary images.
2. In the images, black indicates a binary 1 and white indicates a 0.
3. Operations are limited to binary variables.

TABLE 9.1
The three basic
logical operations.

p	q	p AND q (also $p \cdot q$)	p OR q (also $p + q$)	NOT (p) (also $\bar{p}$)
0	0	0	0	1
0	1	0	1	1
1	0	0	1	0
1	1	1	1	0

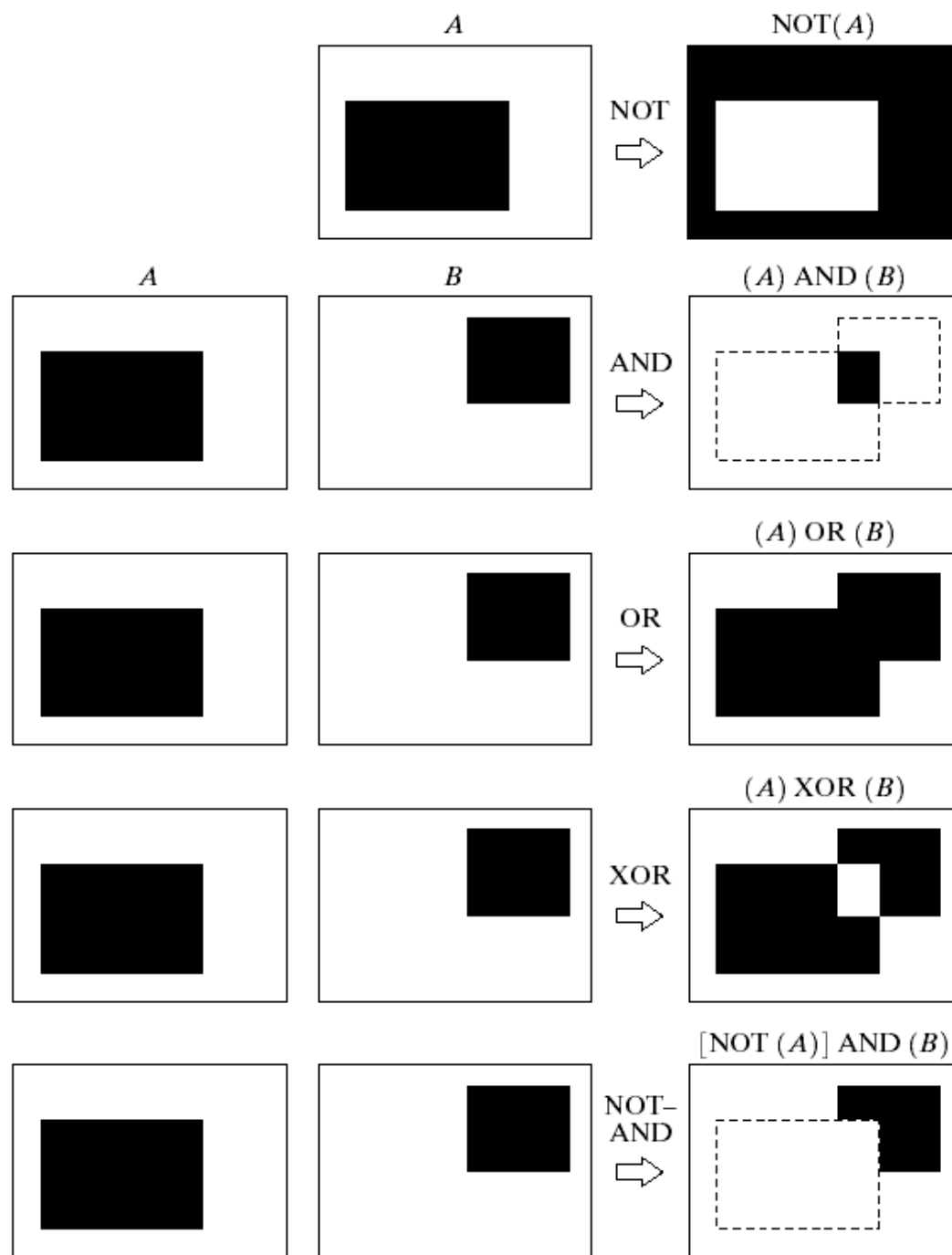


FIGURE 9.3 Some logic operations between binary images. Black represents binary 1s and white binary 0s in this example.

Black = 1
White = 0

Dilation

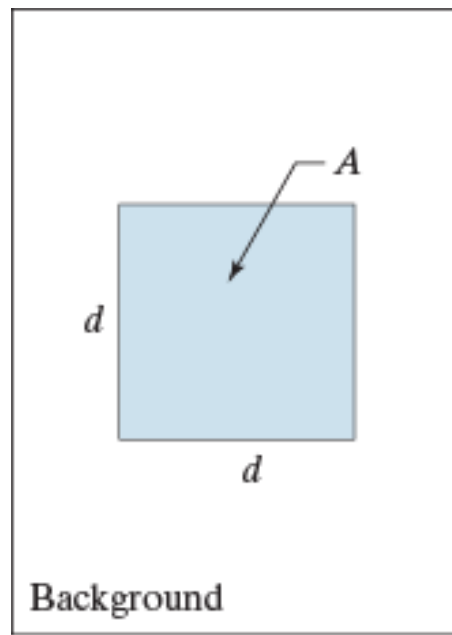
1. Let A and B be sets in Z^2 .
2. The dilation of A by B is defined as

$$A \oplus B = \{z | (\hat{B})_z \cap A \neq \emptyset\}$$

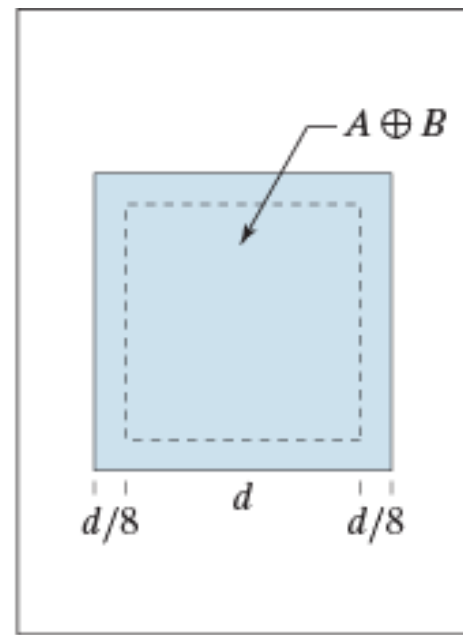
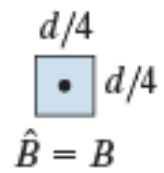
3. The set of all displacements, z , such that $\hat{B}$ and A overlap by at least one element.
4. Another definition.

$$A \oplus B = \{z | [(\hat{B})_z \cap A] \subseteq A\}$$

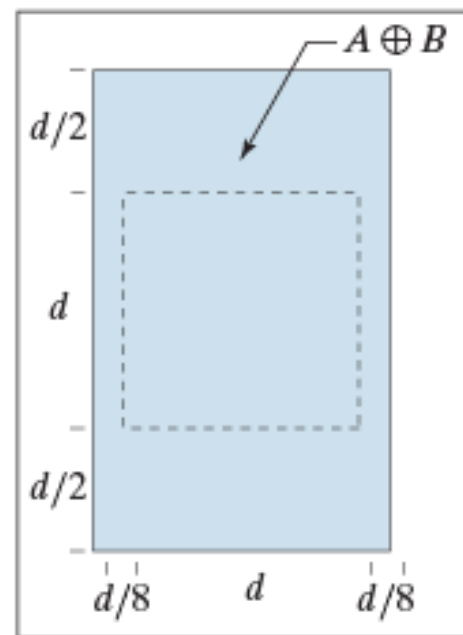
a b c
d e



Image, I



$I \oplus B$



$I \oplus B$

FIGURE 9.6

(a) Image I , composed of set (object) A and background.
 (b) Square SE (the dot is the origin).
 (c) Dilation of A by B (shown shaded).
 (d) Elongated SE.
 (e) Dilation of A by this element. The dotted line in (c) and (e) is the boundary of A , shown for reference.

1. Dashed lines (in Figs. c and e) show the original set for reference.
2. Solid lines (in Figs c and e) show the limit beyond which any further displacements of the origin of $\hat{B}$ by z would cause the intersection of $\hat{B}$ and A to be empty.
3. z is in $A \oplus B$ when A and $\hat{B}$ overlap by at least one element.

Dilation: application

1. Bridging gaps in broken characters

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.

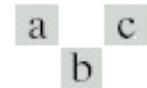
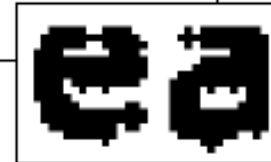


FIGURE 9.5

(a) Sample text of poor resolution with broken characters (magnified view).
(b) Structuring element.
(c) Dilation of (a) by (b). Broken segments were joined.

$$B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Dilation: application

1. Bridging gaps in broken characters

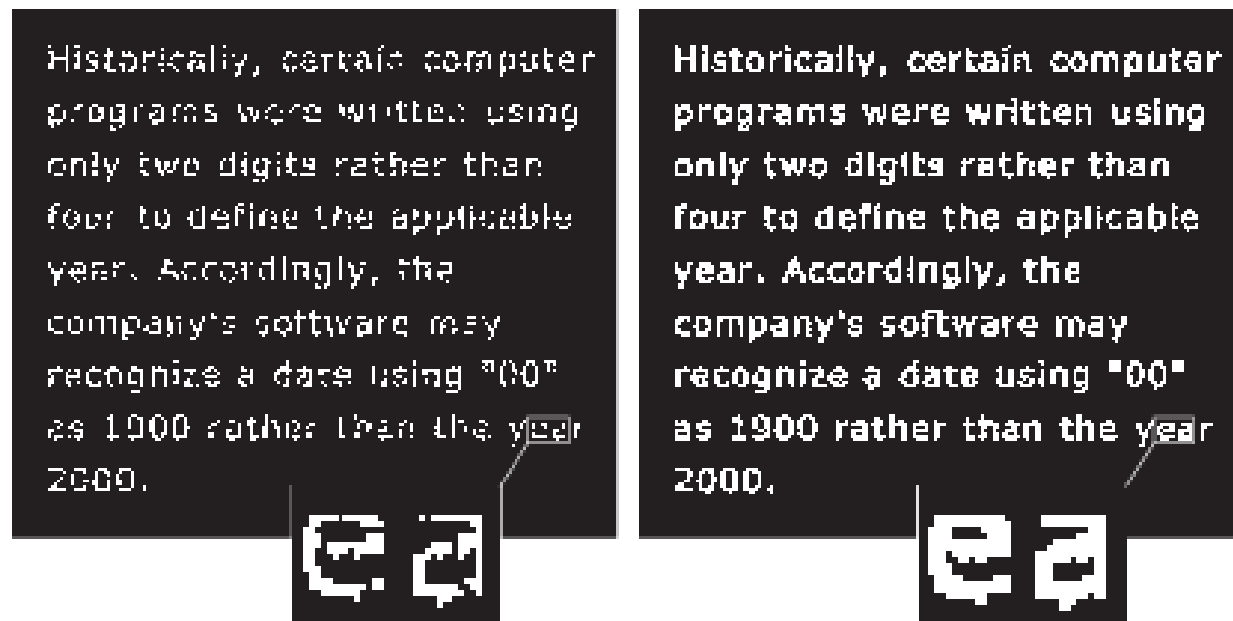


FIGURE 9.7

(a) Low-resolution text showing broken characters (see magnified view).

(b) Structuring element.

(c) Dilation of (a) by (b). Broken segments were joined.



$$B =$$

1	1	1
1	1	1
1	1	1

Erosion

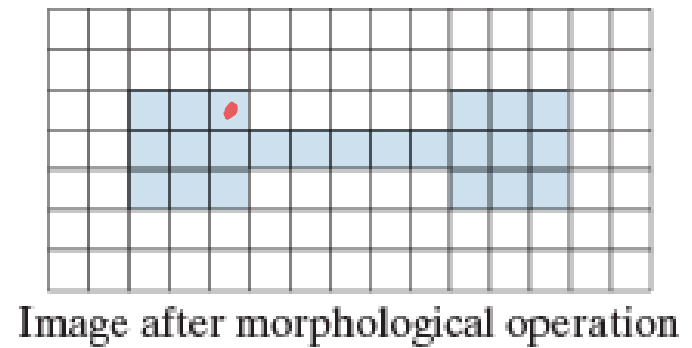
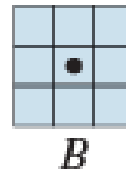
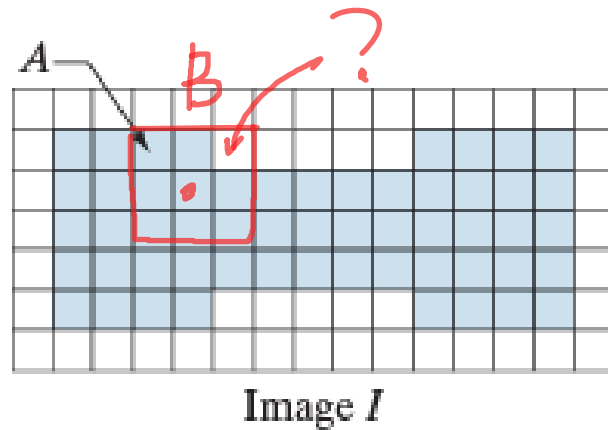
1. Let A and B be sets in Z^2 .
2. The erosion of A by B is defined as

$$A \ominus B = \{z | (B)_z \subseteq A\}$$

3. The set of all displacements, z , such that B , translated by z , is contained in A .
4. Dilation and erosion are duals of each other.

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

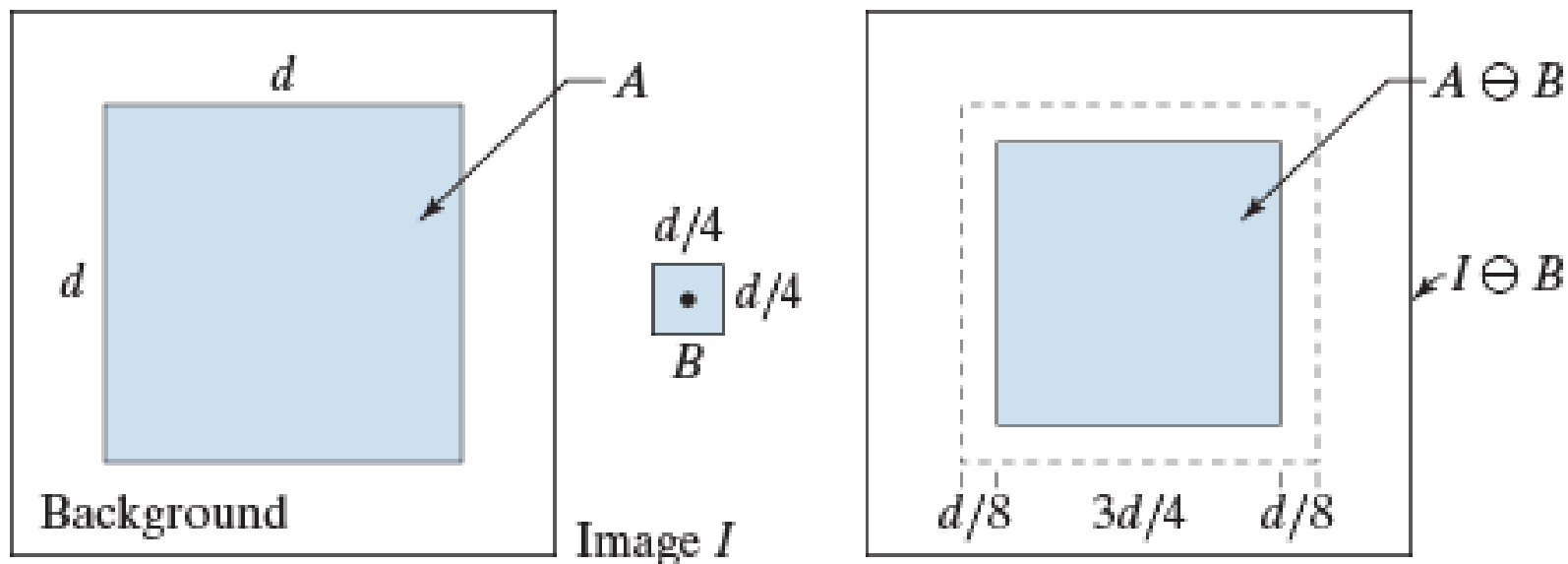
5. Dilation expands objects (represented by ‘1’) in an image and erosion shrinks objects.



a b c

FIGURE 9.3

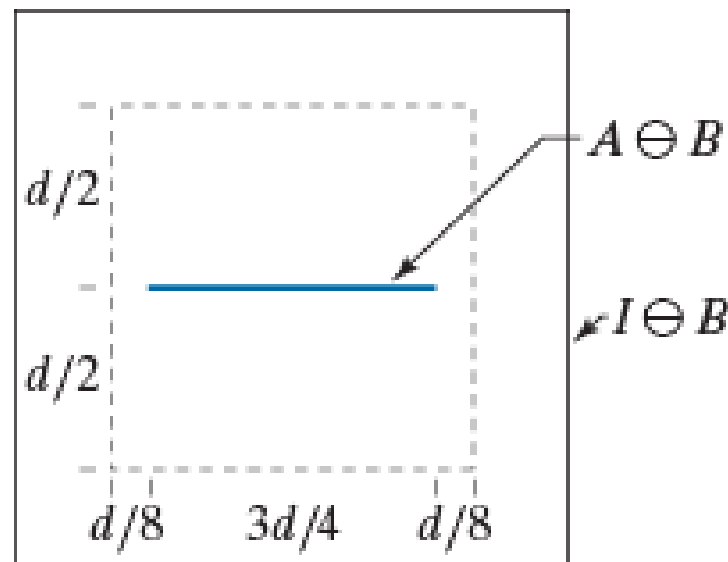
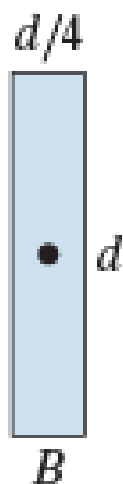
(a) A binary image containing one object (set), A . (b) A structuring element, B . (c) Image resulting from a morphological operation (see text).



a b c
d e

FIGURE 9.4

(a) Image I , consisting of a set (object) A , and background.
 (b) Square SE, B (the dot is the origin).
 (c) Erosion of A by B (shown shaded in the resulting image).
 (d) Elongated SE.
 (e) Erosion of A by B . (The erosion is a line.) The dotted border in (c) and (e) is the boundary of A , shown for reference.



1. Dashed lines (in Figs. c and e) show the original set for reference.
2. Solid lines/shaded region (in Figs. c and e) show the limit beyond which any further displacements of the origin of B by z would cause the erosion set not contained in A .
3. z is in $A \ominus B$ when B is completely contained by the set A .

Erosion: application

1. Eliminating irrelevant details from a binary image.
2. In the images, black indicates a binary 0 and white indicates a 1.
3. All elements in the structuring element have the same binary values as the objects of interest.

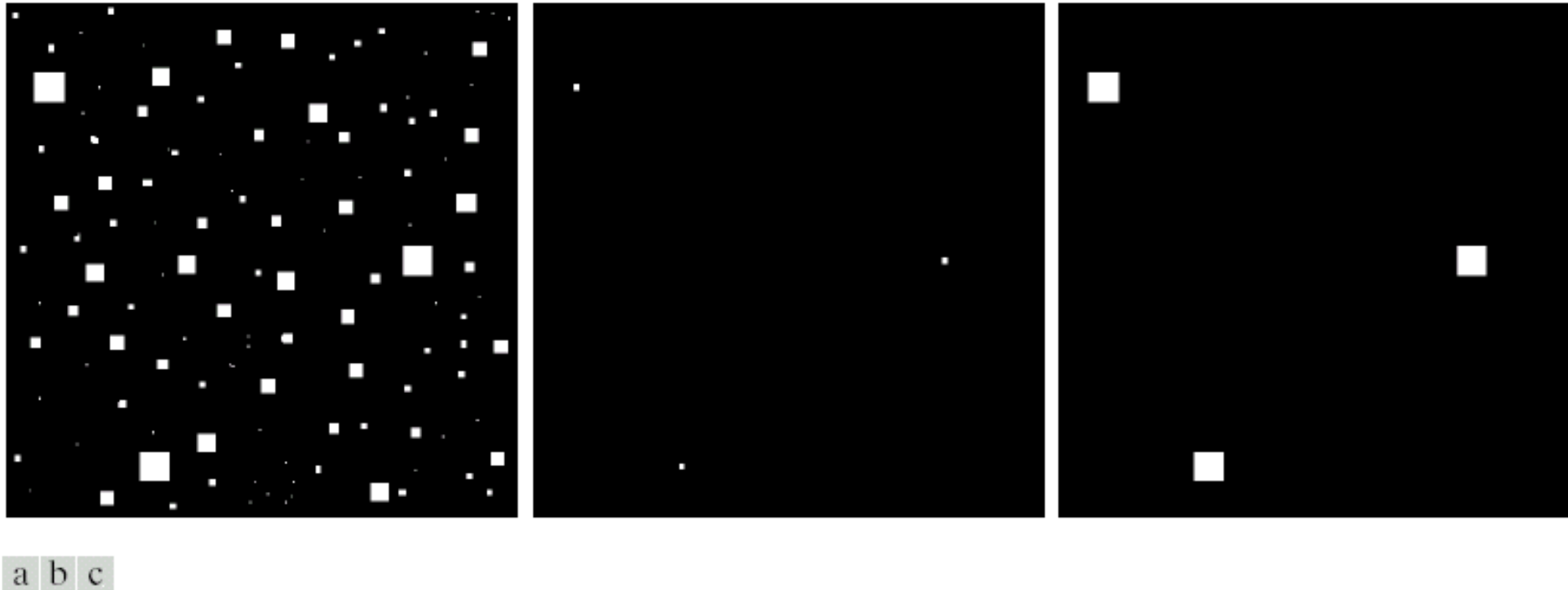


FIGURE 9.7 (a) Image of squares of size 1, 3, 5, 7, 9, and 15 pixels on the side. (b) Erosion of (a) with a square structuring element of 1's, 13 pixels on the side. (c) Dilation of (b) with the same structuring element.

Erosion: application

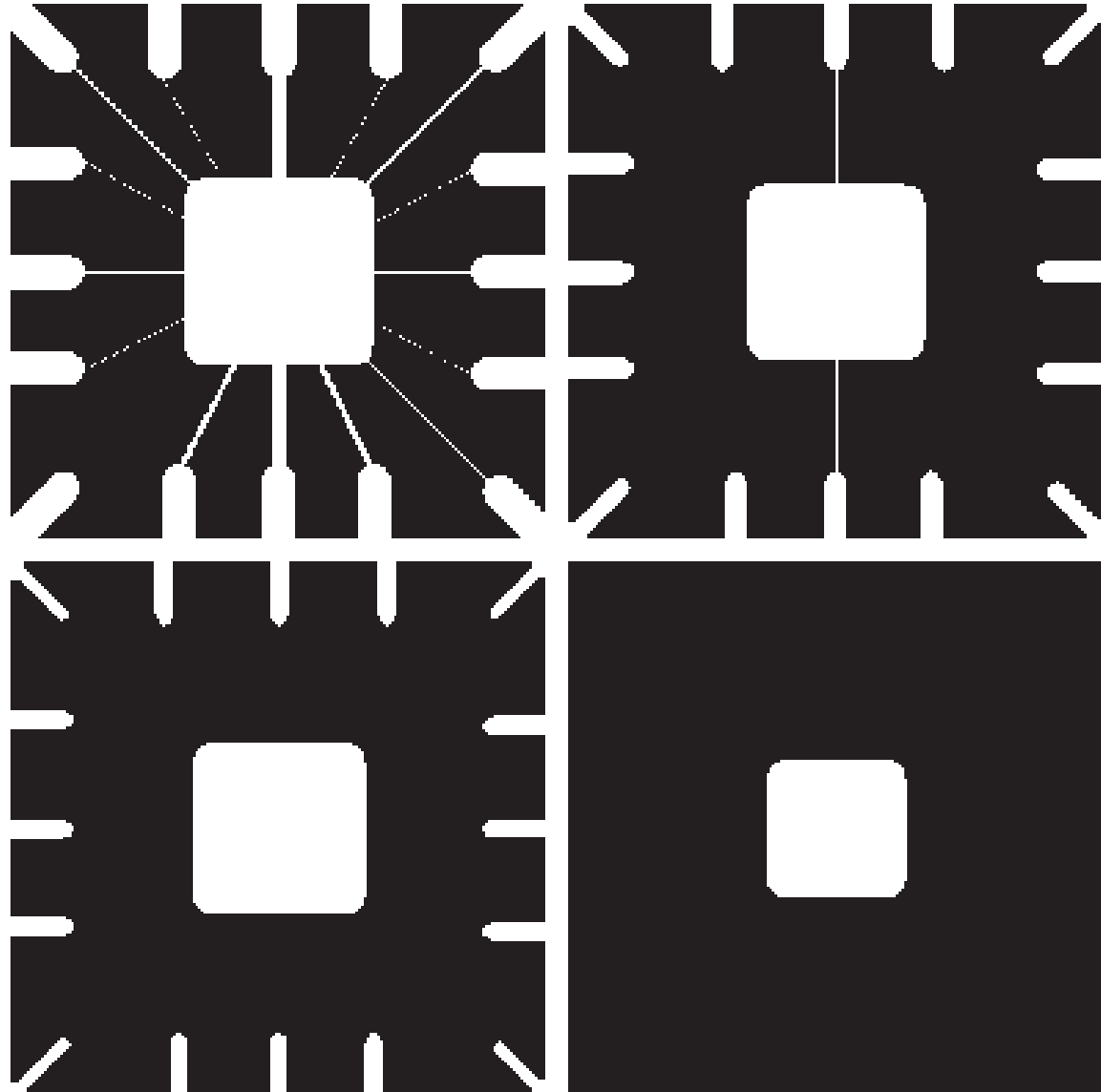
a b
c d

FIGURE 9.5

Using erosion to remove image components.

(a) A 486×486 binary image of a wire-bond mask in which foreground pixels are shown in white.

(b)–(d) Image eroded using square structuring elements of sizes 11×11 , 15×15 , and 45×45 elements, respectively, all valued 1.



Morphological Image Processing

1. We will cover
 - a. Opening and Closing
 - b. Boundary Extraction
 - c. Region Filling
 - d. Connected Components
 - e. Skeletons
 - f. Hit-or-Miss Transform

Opening

1. Recall: Dilation expands an object and erosion shrinks it.
2. In general, opening smooths the object contour by removing corners and regions that cannot be reached by B .

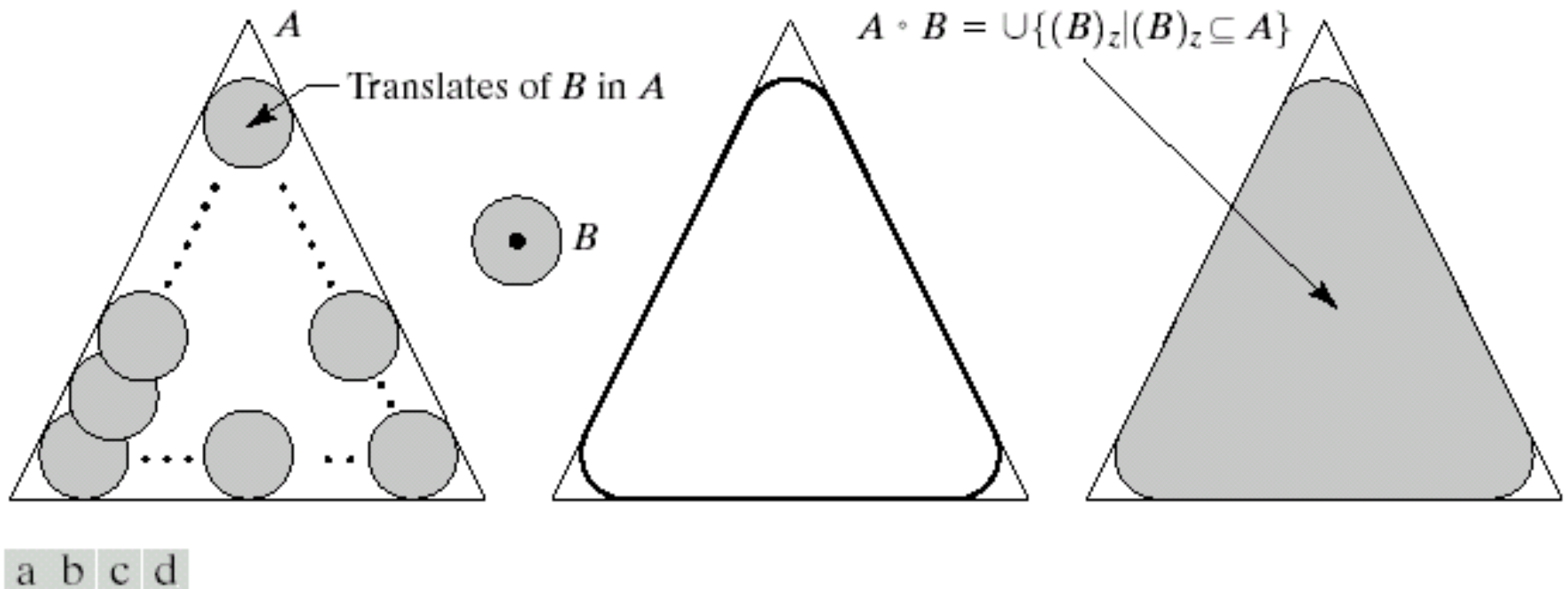


FIGURE 9.8 (a) Structuring element B “rolling” along the inner boundary of A (the dot indicates the origin of B). (c) The heavy line is the outer boundary of the opening. (d) Complete opening (shaded).

Opening

a	b
c	d

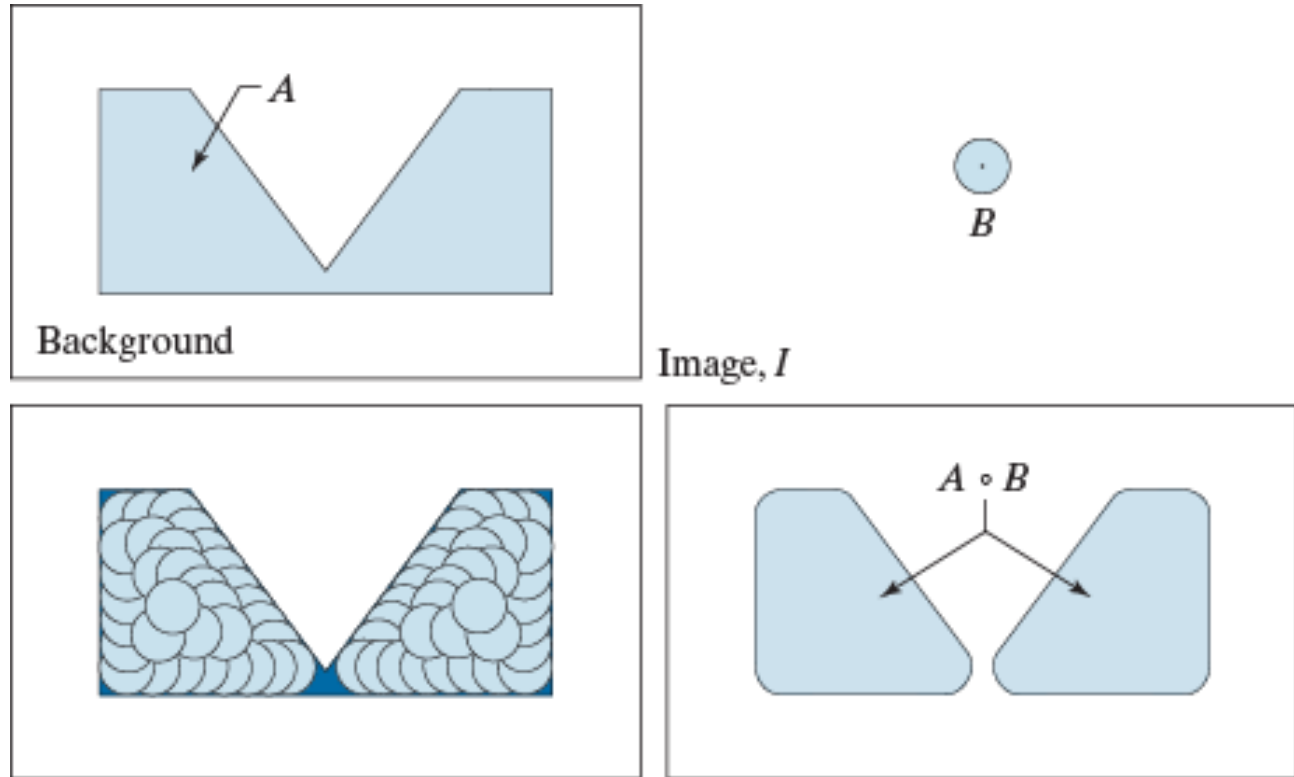
FIGURE 9.8

(a) Image I , composed of set (object) A and background.

(b) Structuring element, B .

(c) Translations of B while being contained in A . (A is shown dark for clarity.)

(d) Opening of A by B .



Opening

3. Definition

$$A \circ B = (A \ominus B) \oplus B$$

- 4. It represents erosion of A by B , followed by dilation of the result by B .
- 5. It means that it takes the union of all translates of B that fit into A . (Imagine you roll the ball inside the object.)
- 6. Another definition

$$A \circ B = \cup \left\{ (B)_z \mid (B)_z \subseteq A \right\}$$

Opening

7. Subimage property

$A \circ B$ is a subset (subimage) of A

8. Convergence property

$$(A \circ B) \circ B = A \circ B$$

Closing

1. In general, closing can also smooth the object contour/boundary by filling the region that cannot be reached by B .

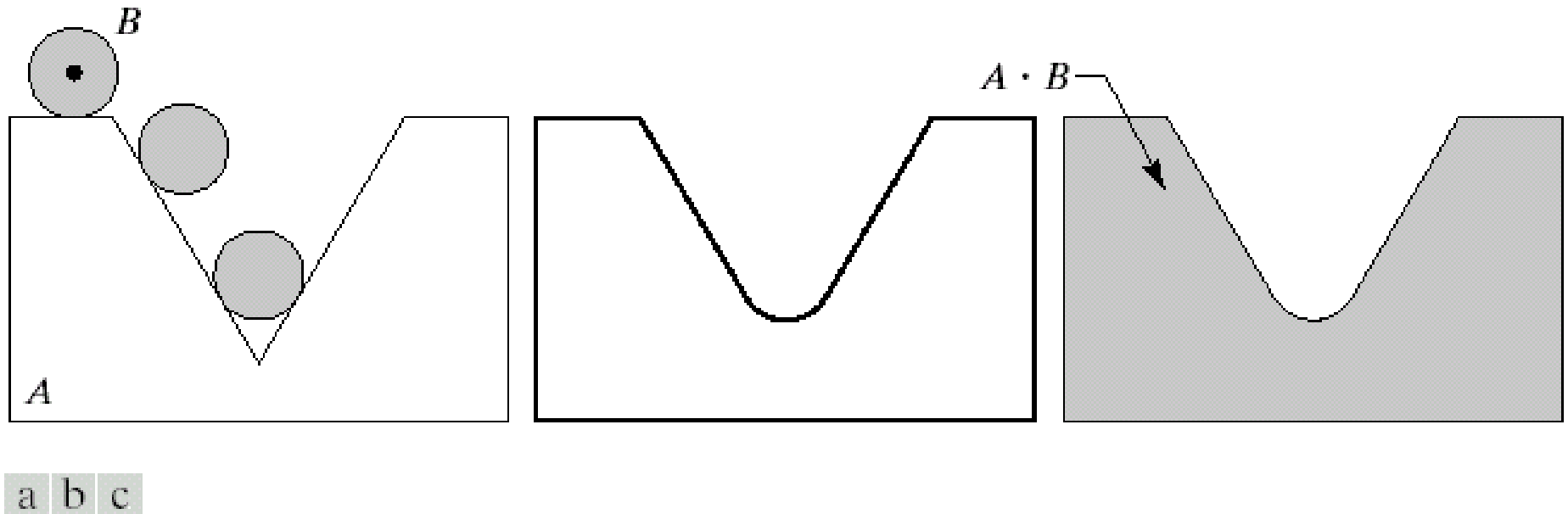
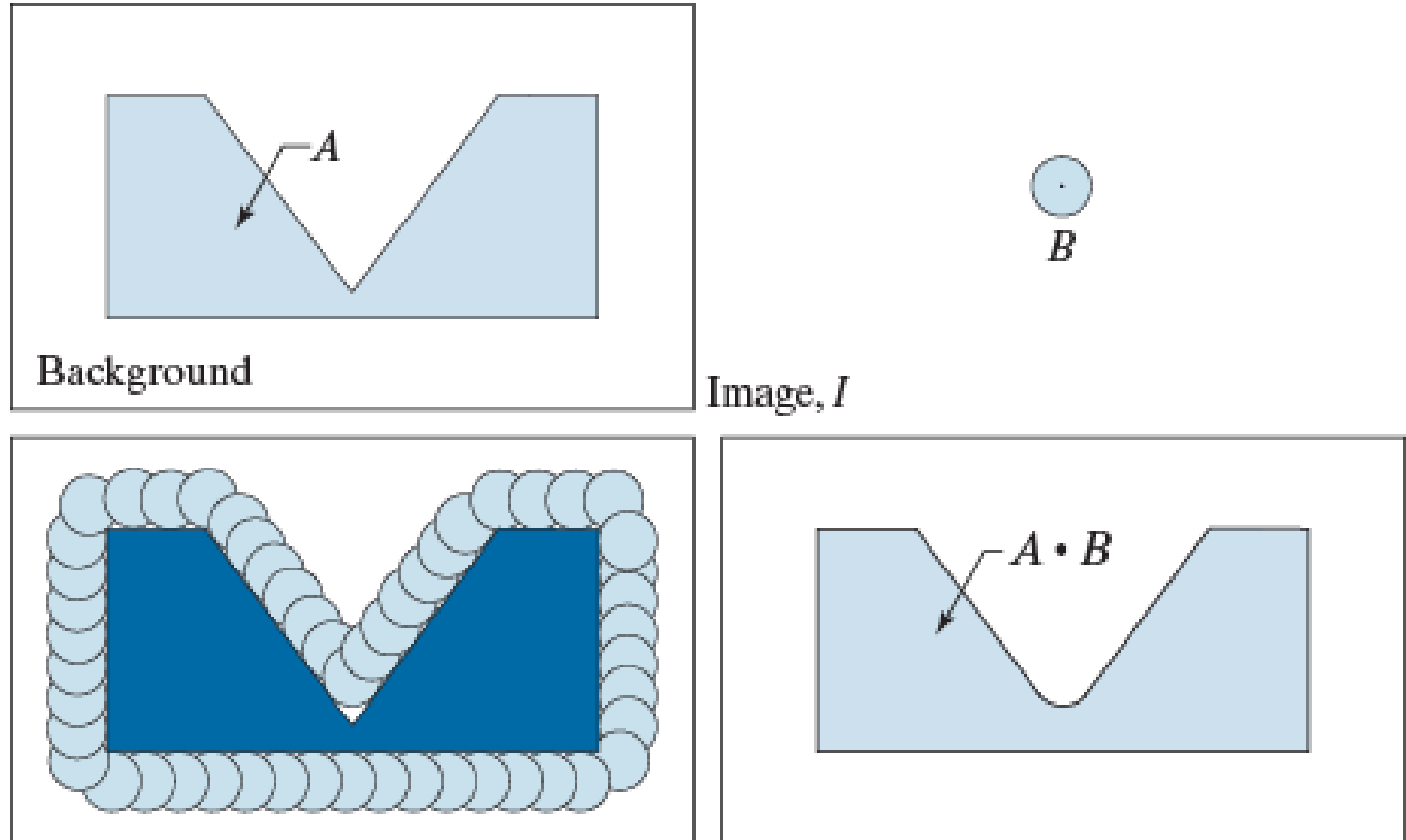


FIGURE 9.9 (a) Structuring element B “rolling” on the outer boundary of set A . (b) Heavy line is the outer boundary of the closing. (c) Complete closing (shaded).

Closing



a b
c d

FIGURE 9.9

(a) Image I , composed of set (object) A , and background.

(b) Structuring element B .

(c) Translations of B such that B does not overlap any part of A . (A is shown dark for clarity.)

(d) Closing of A by B .

Closing

2. Definition

$$A \bullet B = (A \oplus B) \ominus B$$

3. It represents dilation of A by B , followed by erosion of the result by B .
4. Imagine that you roll the ball outside the object instead of rolling the ball inside the object.
5. Opening and closing are dual

$$(A \bullet B)^c = A^c \circ \hat{B}$$

Closing

6. Subimage property

A is a subset (subimage) of $A \bullet B$

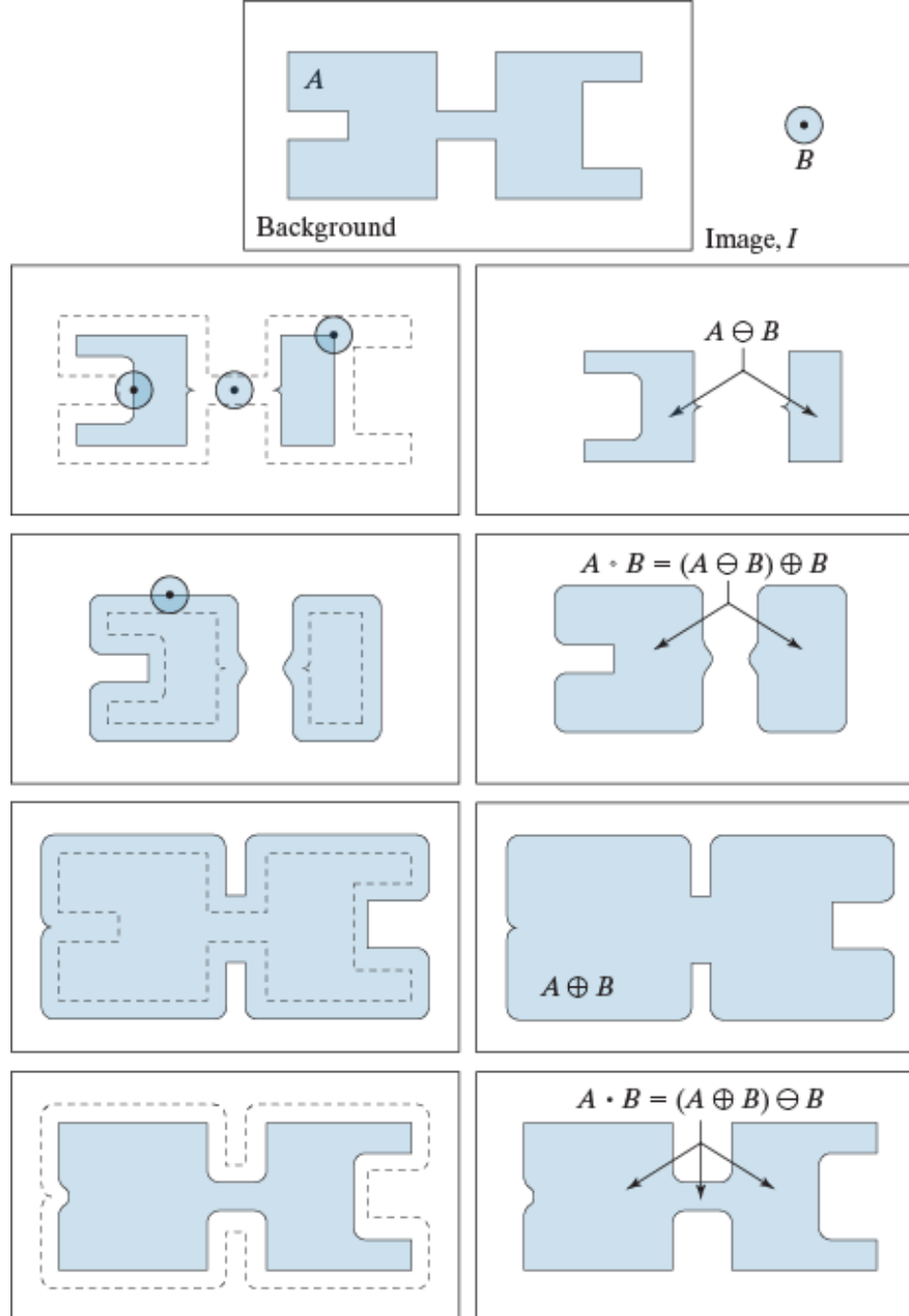
7. Convergence property

$$(A \bullet B) \bullet B = A \bullet B$$

a
b c
d e
f g
h i

FIGURE 9.10

Morphological opening and closing.
(a) Image I , composed of a set (object) A and background; a solid, circular structuring element is shown also. (The dot is the origin.)
(b) Structuring element in various positions.
(c)-(i) The morphological operations used to obtain the opening and closing.



Opening results
of Figure (a)

Closing results
of Figure (a)

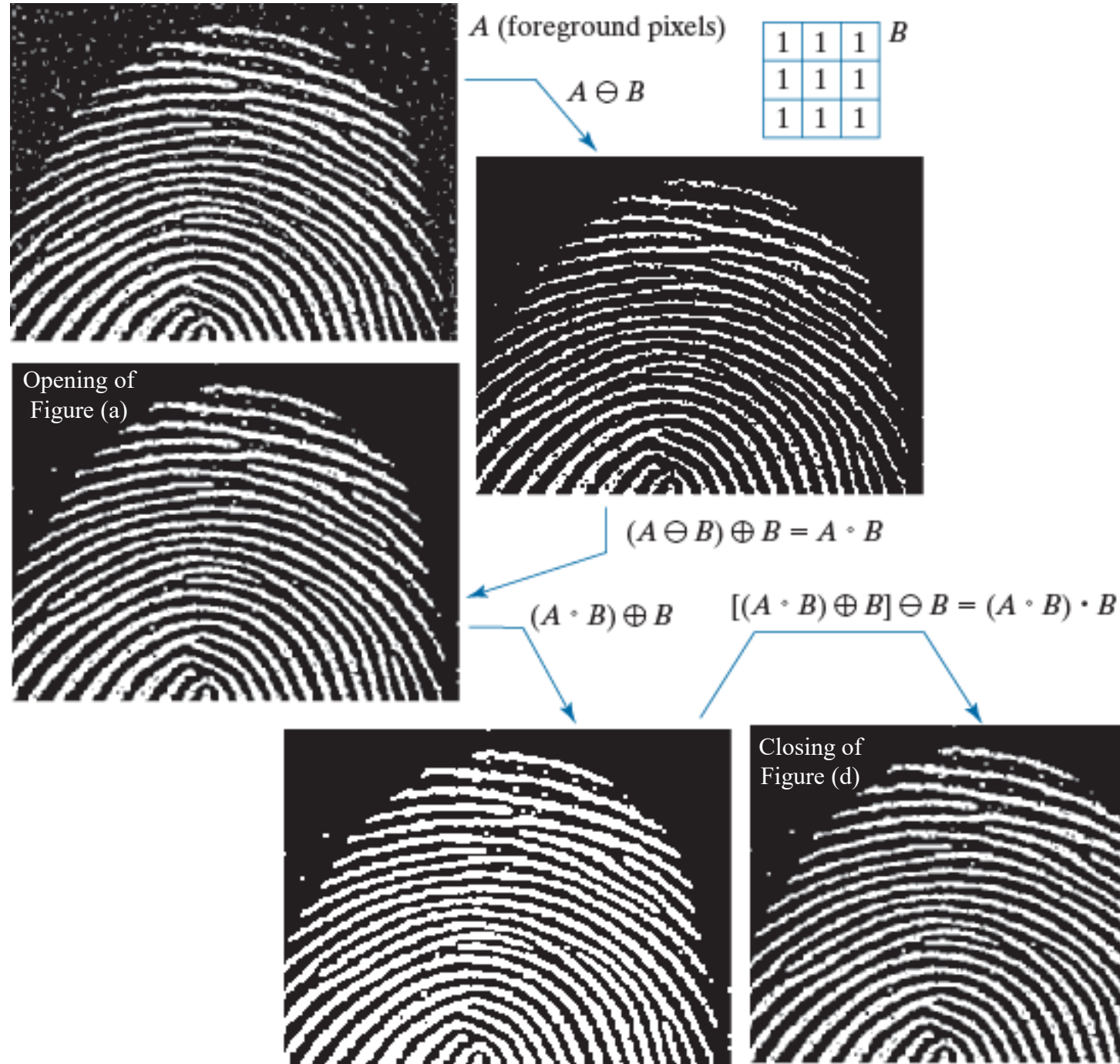


FIGURE 9.11

(a) Noisy image.
 (b) Structuring element.
 (c) Eroded image.
 (d) Dilation of the erosion (opening of A). (e) Dilation of the opening.
 (f) Closing of the opening.
 (Original image courtesy of the National Institute of Standards and Technology.)

Boundary Extraction

1. We define

$\beta(A)$ denotes the boundary of a set A

2. The boundary is extracted by using

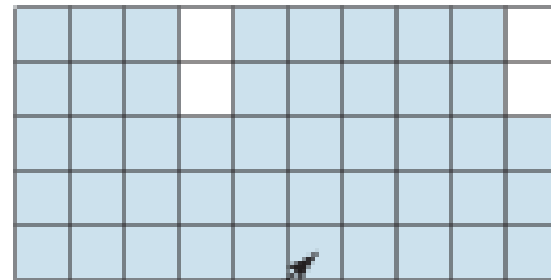
$$\beta(A) = A - (A \ominus B)$$

3. It means the boundary is the difference between the object and the ‘eroded’ object.

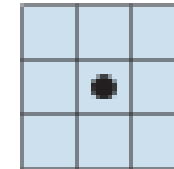
a	b
c	d

FIGURE 9.15

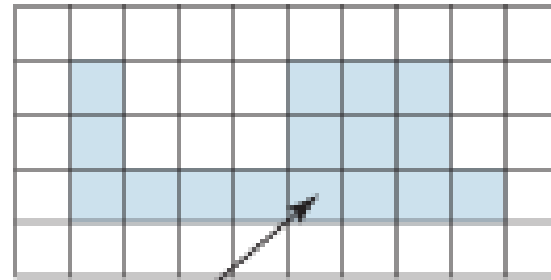
- (a) Set, A , of foreground pixels.
 (b) Structuring element.
 (c) A eroded by B .
 (d) Boundary of A .



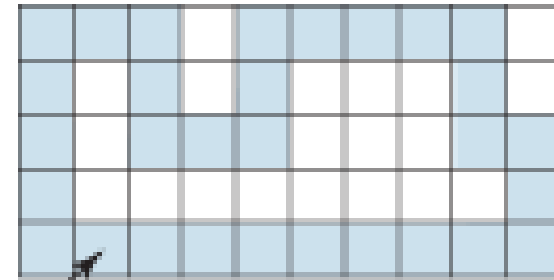
A



B



$A \ominus B$



$\beta(A) = A - (A \ominus B)$

Shaded region = 1, white region = 0



a b

FIGURE 9.14

(a) A simple binary image, with 1's represented in white. (b) Result of using Eq. (9.5-1) with the structuring element in Fig. 9.13(b).

Region Filling

1. Beginning with a point p inside the region boundary, the objective is to fill the bounded region with 1's.
2. The algorithm is defined as follows.

$$X_k = (X_{k-1} \oplus B) \cap A^c$$

3. $k = 1, 2, 3, \dots$
4. $X_0 = p$
5. B is the structuring element.
6. Intersection with A^c limits the result to inside the region of interest.
7. The algorithm stops when $X_k = X_{k-1}$. This means the algorithm stops when there is no change in the region size.

a	b	c
d	e	f
g	h	i

FIGURE 9.17

Hole filling.

(a) Set A (shown shaded) contained in image I .

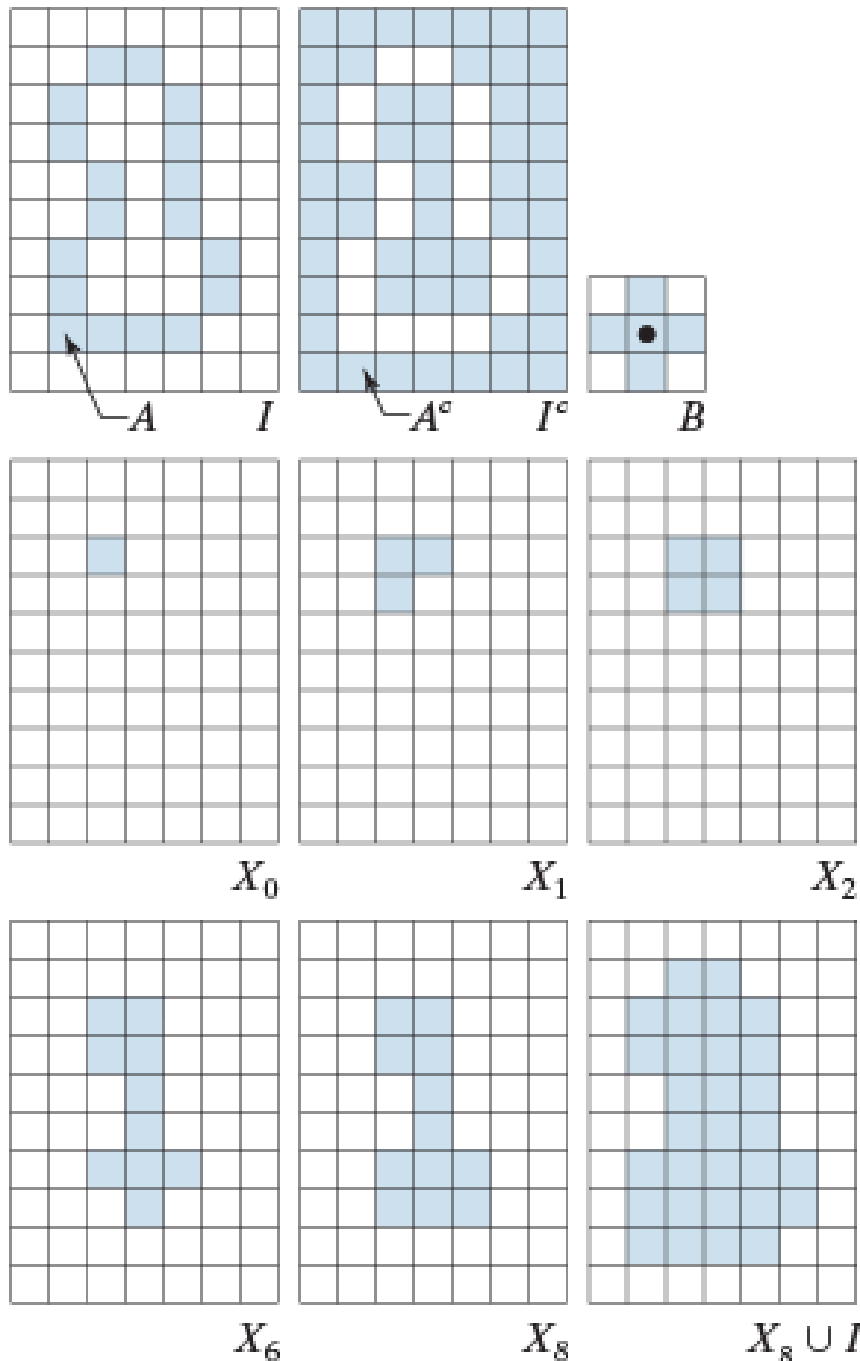
(b) Complement of I .

(c) Structuring element B . Only the foreground elements are used in computations

(d) Initial point inside hole, set to 1.

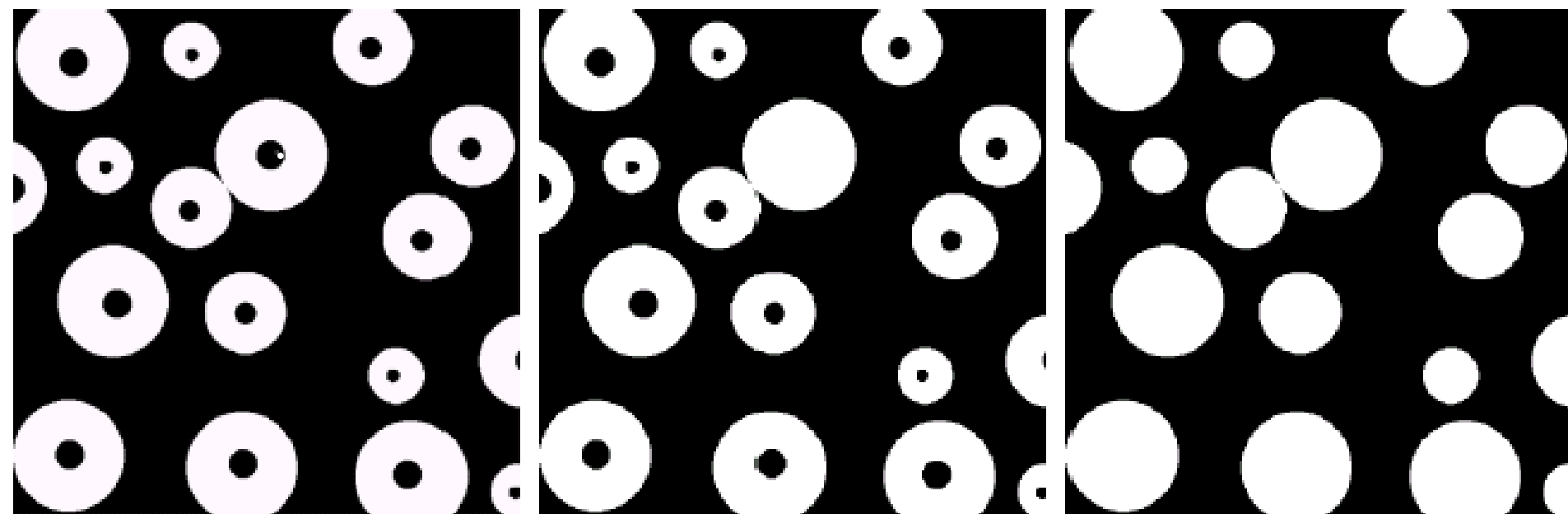
(e)–(h) Various steps of Eq. (9-19).

(i) Final result [union of (a) and (h)].



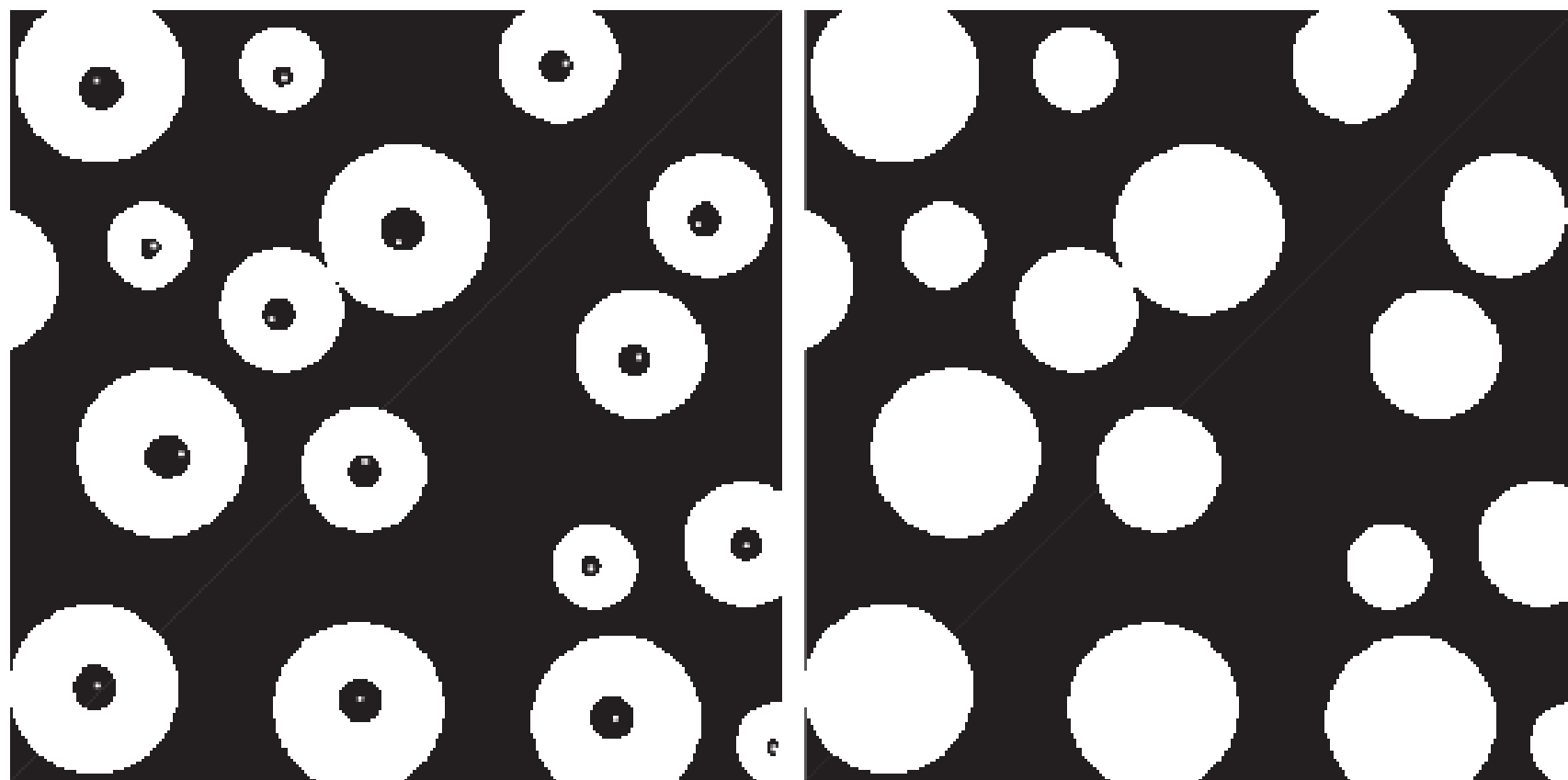
Shaded region = 1

White region = 0



a b c

FIGURE 9.16 (a) Binary image (the white dot inside one of the regions is the starting point for the region-filling algorithm). (b) Result of filling that region (c) Result of filling all regions.



a b

FIGURE 9.18

(a) Binary image. The white dots inside the regions (shown enlarged for clarity) are the starting points for the hole-filling algorithm.
(b) Result of filling all holes.

Connected Components

1. Similar to region filling, the procedure is defined as follows.

$$X_k = (X_{k-1} \oplus B) \cap A$$

2. Beginning with a point p inside the object, the objective is to fill out the entire object with 1's.
3. The intersection with A at each iteration eliminates dilations centred on elements labelled 0.

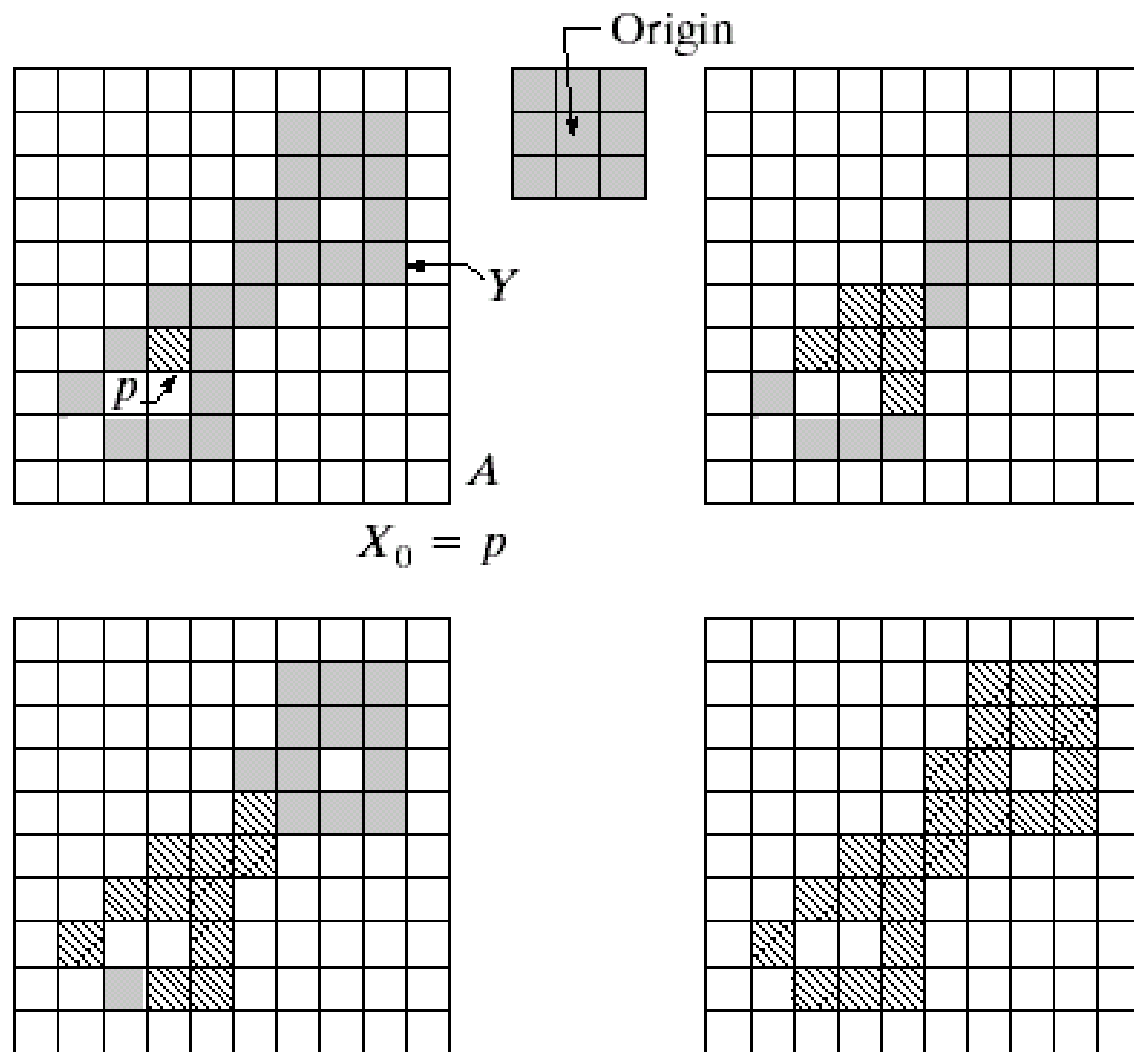
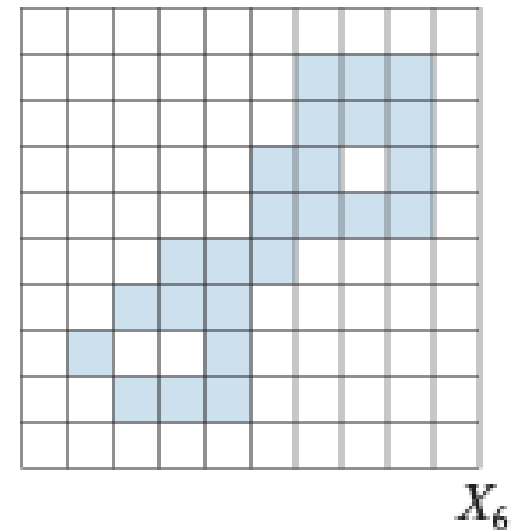
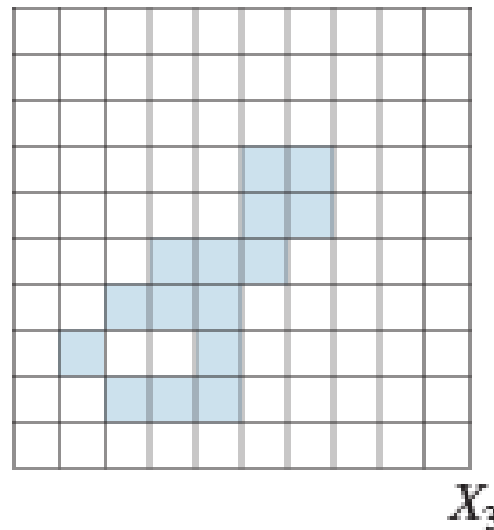
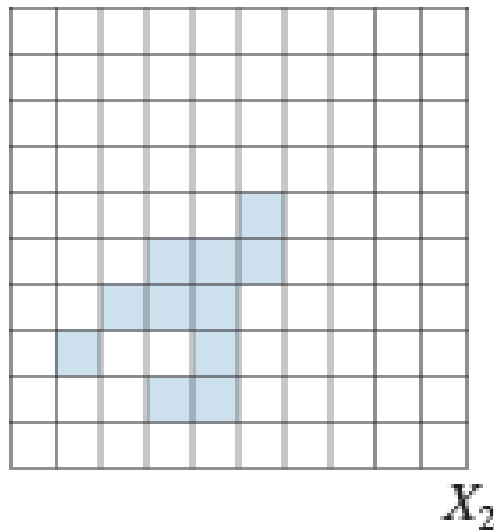
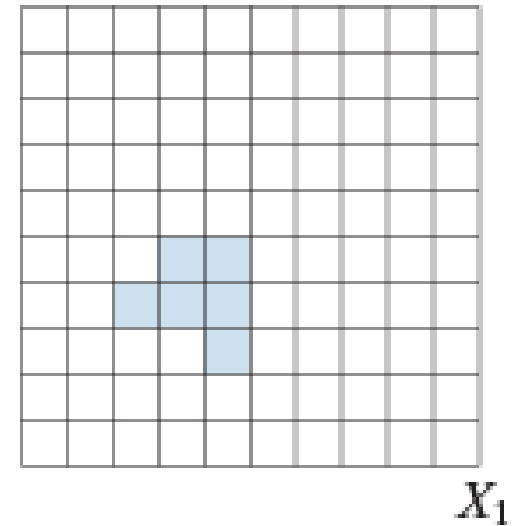
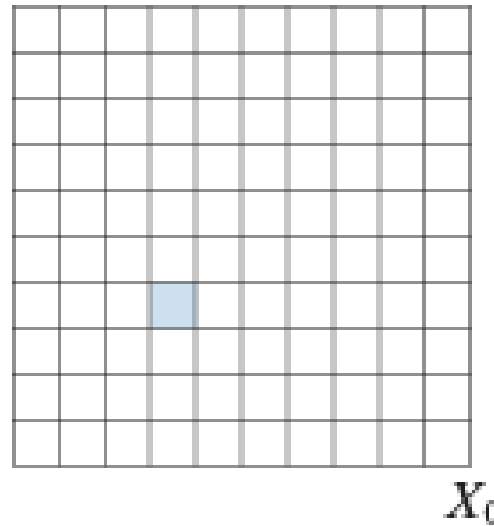
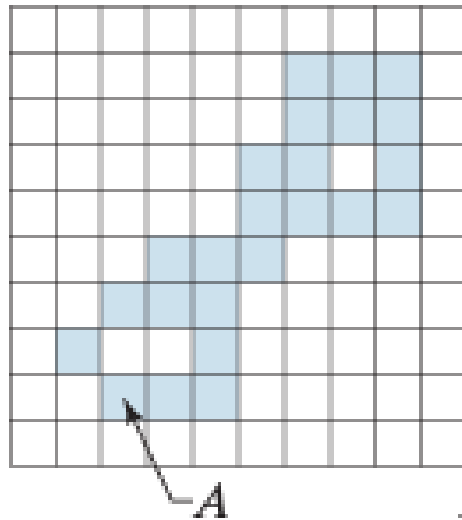
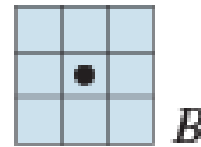


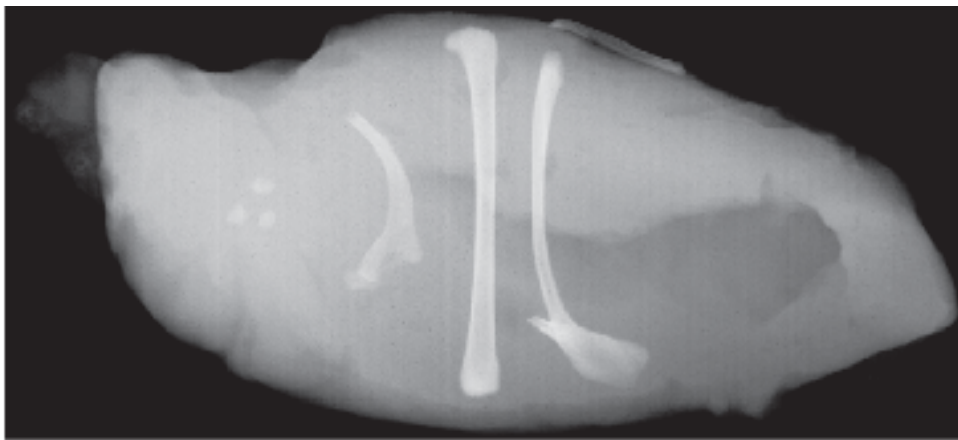
FIGURE 9.17 (a) Set A showing initial point p (all shaded points are valued 1, but are shown different from p to indicate that they have not yet been found by the algorithm). (b) Structuring element. (c) Result of first iterative step. (d) Result of second step. (e) Final result.



Shaded region = 1
White region = 0

a
b c d
e f g

FIGURE 9.19
(a) Structuring element.
(b) Image containing a set with one connected component.
(c) Initial array containing a 1 in the region of the connected component.
(d)–(g) Various steps in the iteration of Eq. (9-20)



a
b
c d

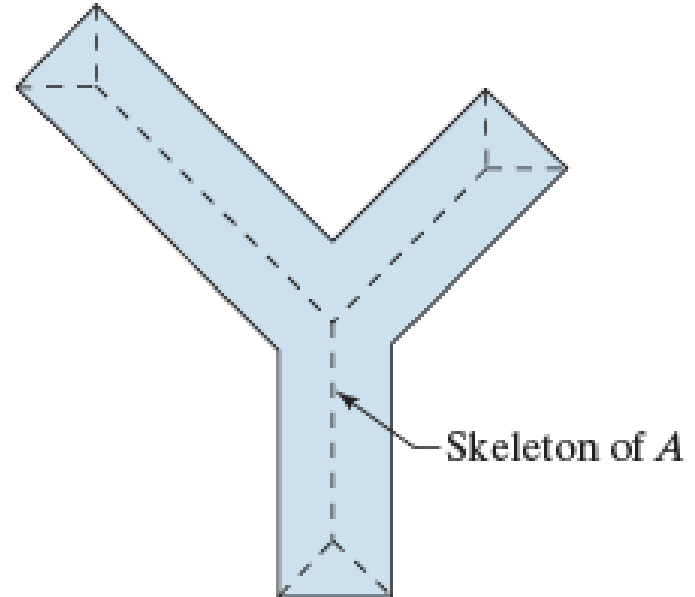
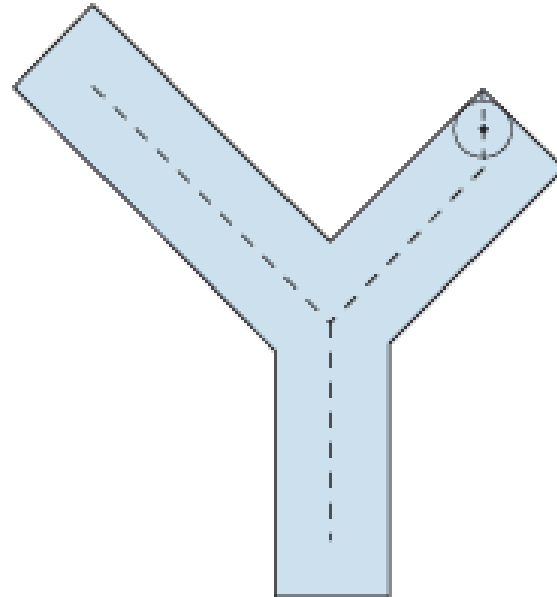
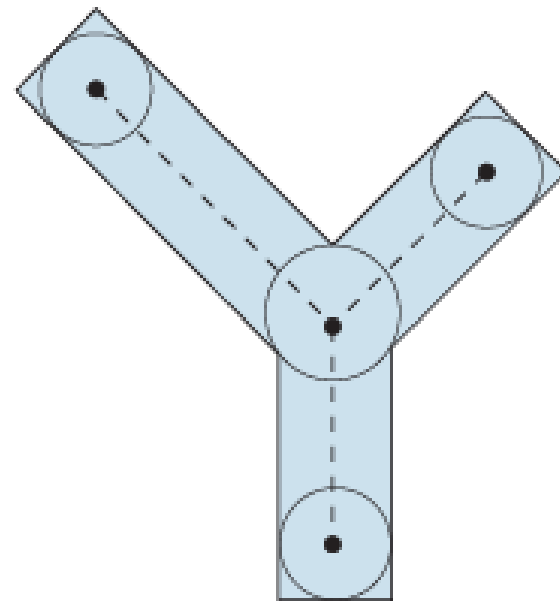
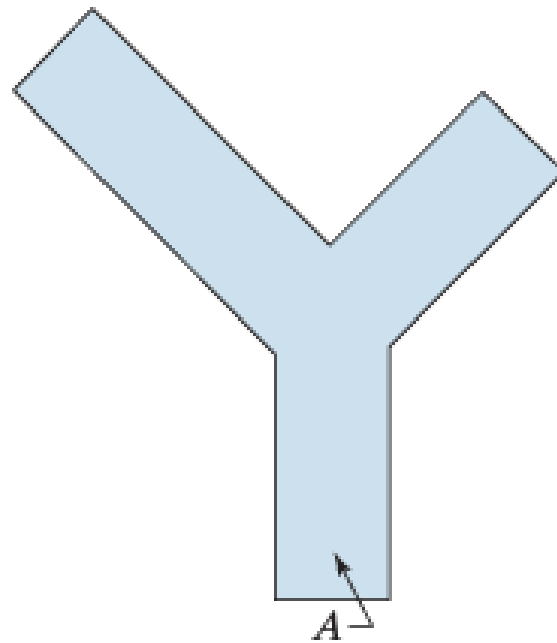


FIGURE 9.20
 (a) X-ray image of a chicken filet with bone fragments.
 (b) Thresholded image (shown as the negative for clarity).
 (c) Image eroded with a 5×5 SE of 1's.
 (d) Number of pixels in the connected components of (c). (Image (a) courtesy of NTB Elektronische Geraete GmbH, Diepholz, Germany, www.ntbxray.com.)

Connected component	No. of pixels in connected comp
01	11
02	9
03	9
04	39
05	133
06	1
07	1
08	743
09	7
10	11
11	11
12	9
13	9
14	674
15	85

Skeletons

1. Definition of a skeleton, $S(A)$, of a set A is given as follows
 - a. If
 - i. z is a point of $S(A)$ and
 - ii. $(D)_z$ is the largest disk centred at z and contained in A ,
 - b. then
 - i. one cannot find a larger disk (not necessarily centred at z) containing $(D)_z$ and included in A , and
 - ii. the disk $(D)_z$ is called *maximum disk*.
 - iii. the disk $(D)_z$ touches the boundary of A at two or more different places.



a	b
c	d

FIGURE 9.25

- (a) Set A .
- (b) Various positions of maximum disks whose centers partially define the skeleton of A .
- (c) Another maximum disk, whose center defines a different segment of the skeleton of A .
- (d) Complete skeleton (dashed).

2. The algorithm is given by $S(A)$ is the union of skeleton subsets $S_k(A)$

$$S(A) = \bigcup_{k=0}^K S_k(A)$$

3. with the skeleton subset defined by

$$S_k(A) = (A \ominus kB) - (A \ominus kB) \circ B$$

4. where B is a structuring element, and

$$(A \ominus kB) = (\dots (A \ominus B) \ominus B) \ominus \dots) \ominus B$$

k successive erosions of A by B

5. K is the last iterative step before A erodes to an empty set.

$$K = \max\{k | (A \ominus kB) \neq \emptyset\}$$

Reconstruction of skeletons

1. The original set A can be reconstructed by using the equation

$$A = \bigcup_{k=0}^K (S_k(A) \oplus kB)$$

2. where

$$(S_k(A) \oplus kB) = \underbrace{\left(\left(\left((S_k(A) \oplus B) \oplus B \right) \oplus B \right) \oplus \dots \right) \oplus B}_{k \text{ successive dilations}}$$

k successive dilations of $S_k(A)$

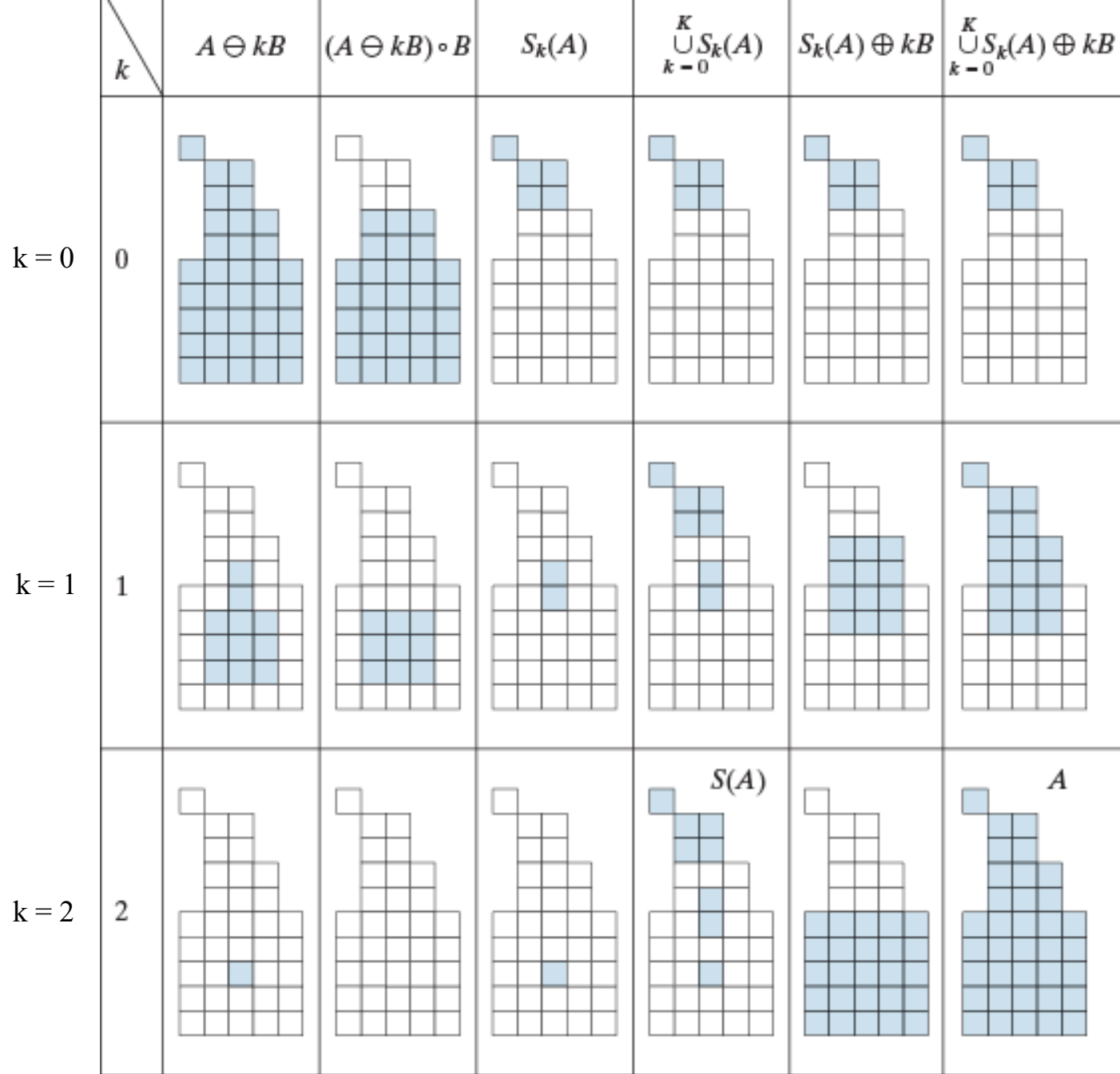
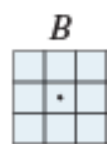


FIGURE 9.26 Implementation of Eqs. (9-28) through (9-33). The original set is at the top left, and its morphological skeleton is at the bottom of the fourth column. The reconstructed set is at the bottom of the sixth column.

Shaded region = 1
White region = 0



Hit-or-Miss Transform

1. Hit-or-Miss Transform (HMT) is a basic tool for shape detection.
2. HMT uses two structuring elements: B_1 and B_2 .
3. HMT of image I is defined as

$$\begin{aligned} I \circledast B_{1,2} &= \{z | (B_1)_z \subseteq A \text{ and } (B_2)_z \subseteq A^c\} \\ &= (A \ominus B_1) \cap (A^c \ominus B_2) \end{aligned}$$

4. In HMT, B_1 is contained in A and, simultaneously, B_2 is contained in A^c . The translation z is the same translation of both structuring elements.
5. “Miss” in HMT = B_2 finds a match in A^c is the same as B_2 not finding a match in A .

Hit-or-Miss Transform

6. Suppose that we want to find the location of the origin of the object (set) D , which is a subset of A .
7. The two structuring elements, B_1 and B_2 , are designed for detecting the properties of both the foreground and background pixels, as the object shape is determined by their geometrical arrangements.
8. In Figure 9.12(c), the structuring element B_2 is designed to detect the border of D .
9. In Figure 9.12(f), the interaction of the outputs (d) and (e) gives the location of the origin of the object (set) D .

a	b
c	d
e	f

FIGURE 9.12

(a) Image consisting of a foreground (1's) equal to the union, A , of set of objects, and a background of 0's.

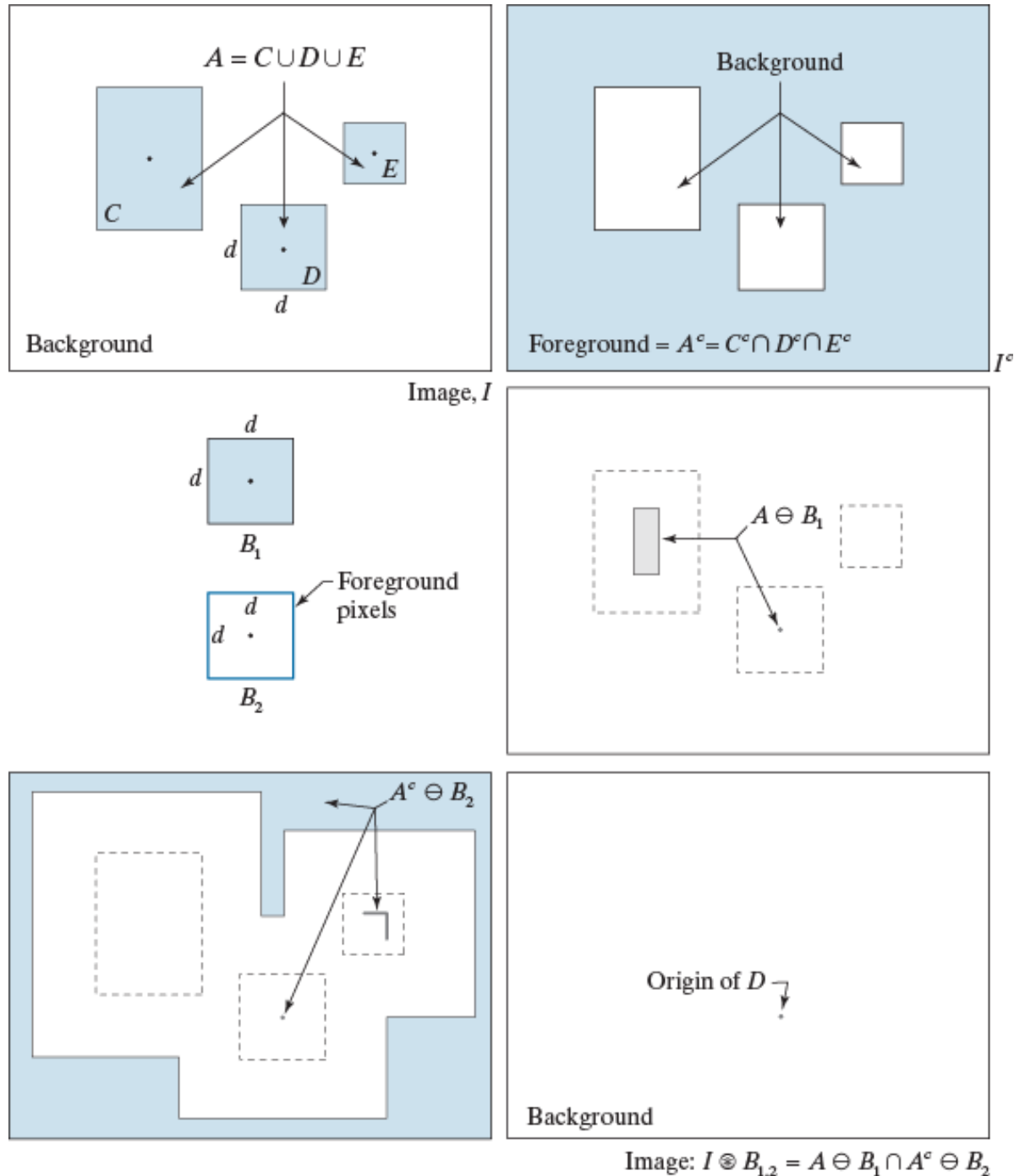
(b) Image with its foreground defined as A^c .

(c) Structuring elements designed to detect object D .

(d) Erosion of A by B_1 .

(e) Erosion of A^c by B_2 .

(f) Intersection of (d) and (e), showing the location of the origin of D , as desired. The dots indicate the origin of their respective components. Each dot is a single pixel.



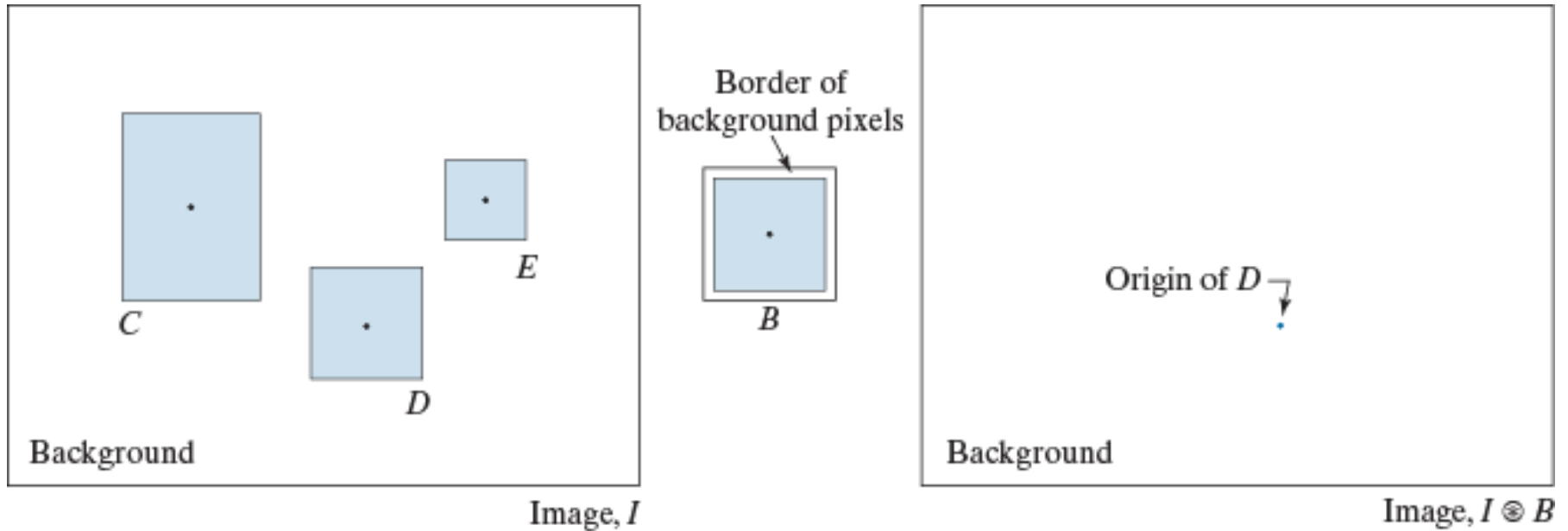
Hit-or-Miss Transform

1. In Figure 9.13, it shows that the detection can be done by using one structuring element B alone.
2. In Figure 9.13(b), the structuring element is the same as D but has a border of background elements with a width of one pixel. Now HMT becomes

$$I \odot B_{1,2} = \{z | (B)_z \subseteq I\}$$

3. We need to check if B_z is a subset of image I , which has both foreground and background pixels.
4. In Figure 9.13(c), the output is the same as the output shown in Figure 9.12(f). However, only one structuring element is used.

Hit-or-Miss Transform



a b c

FIGURE 9.13 Same solution as in Fig. 9.12, but using Eq. (9-17) with a single structuring element.

Hit-or-Miss Transform

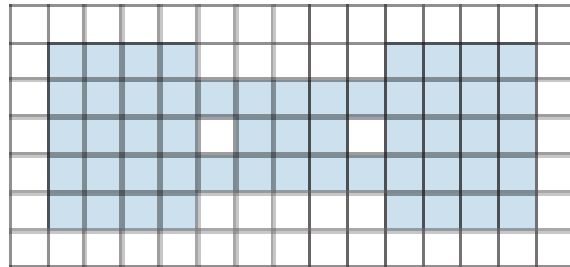
1. In Figure 9.14(a), it shows the result of using a small structuring element of foreground and background elements, which is designed for detecting one-pixel holes.
2. In Figure 9.14(b), it is designed for detecting the foreground corner pixel of the top, right corner of the object in I .
3. In Figure 9.14(c), the structuring element has foreground, background, and “don’t care” (x) elements. The use of “don’t care” elements increases detection flexibility.

Hit-or-Miss Transform

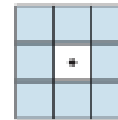
a	b	c
d	e	f
g	h	i

FIGURE 9.14

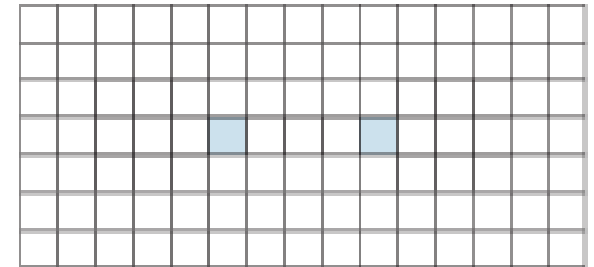
Three examples of using a single structuring element and Eq. (9-17) to detect specific features. First row: detection of single-pixel holes. Second row: detection of an upper-right corner. Third row: detection of multiple features.



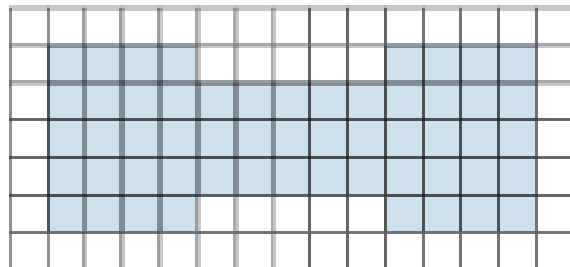
Image, I



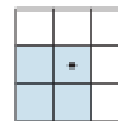
B



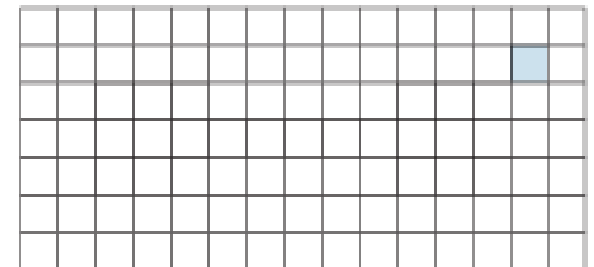
Image, $I \otimes B$



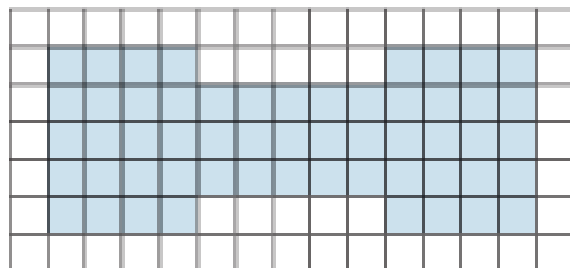
Image, I



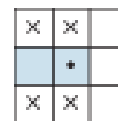
B



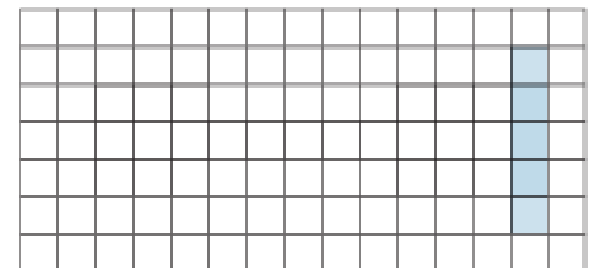
Image, $I \otimes B$



Image, I



B



Image, $I \otimes B$

Morphological Image Processing for Gray-Scale Images

Morphological Image Processing

1. Extensions to gray-scale images.
 - a. Dilation
 - b. Erosion
 - c. Opening
 - d. Closing
2. Some applications.

Notations

1. Let $f(x,y)$ be the input image.
2. Let $b(x,y)$ be the structuring element.
3. (x,y) = position coordinates.
4. $f(x,y)$ and $b(x,y)$ are functions that assign a gray-level value to each position located at (x,y) .

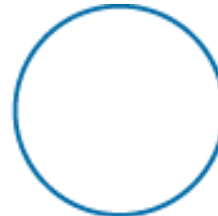


FIGURE 9.36

Nonflat and flat structuring elements, and corresponding horizontal intensity profiles through their centers. All examples in this section are based on flat SEs.



Nonflat SE



Flat SE



Intensity profile



Intensity profile

Dilation

1. 2D gray-scale dilation of f by b is defined as

$$(f \oplus b)(s, t) = \max\{f(s - x, t - y) + b(x, y) \mid (s - x, t - y) \in D_f; (x, y) \in D_b\}$$

2. D_f and D_b are the domains of $f(s-x, t-y)$ and $b(x, y)$.
3. If the function b is zero, the gray-level dilation operation is the same as finding the local maximum.
4. $(s-x)$ and $(t-y)$ have to be in the domain of $f(s-x, t-y)$.
5. x and y have to be in the domain of $b(x, y)$.
6. Points 3 and 4 are similar to the conditions in the binary definition of dilation, which make sure that *the two sets have to overlap by at least one element*.

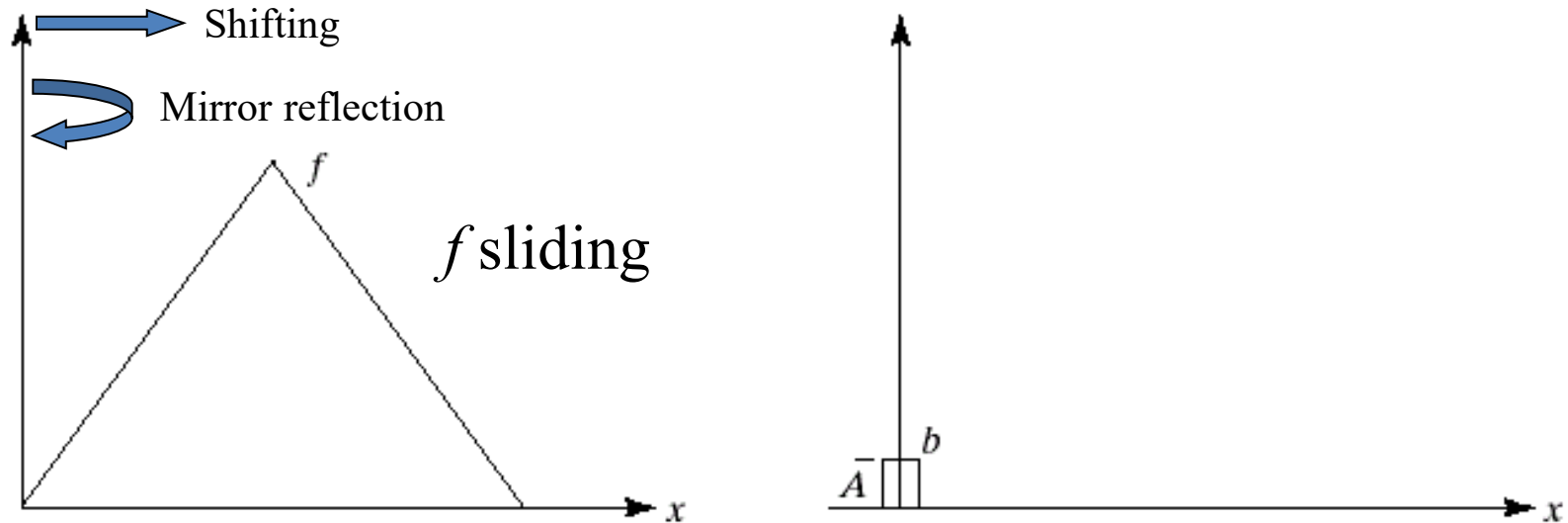
Dilation

7. 1D gray-scale dilation of f by b is defined as

$$(f \oplus b)(s) = \max\{f(s - x) + b(x) | (s - x) \in D_f; x \in D_b\}$$

8. $f(-x)$ is a mirror reflection of $f(x)$.

9. $f(s-x)$ is equivalent to shifting to the right (left) when the value of s is positive (negative).



Dilation

10. As f sliding by b is no different with b sliding by f , the actual mechanics of gray-scale dilation are easier to visualize if b is the function that slides past f .

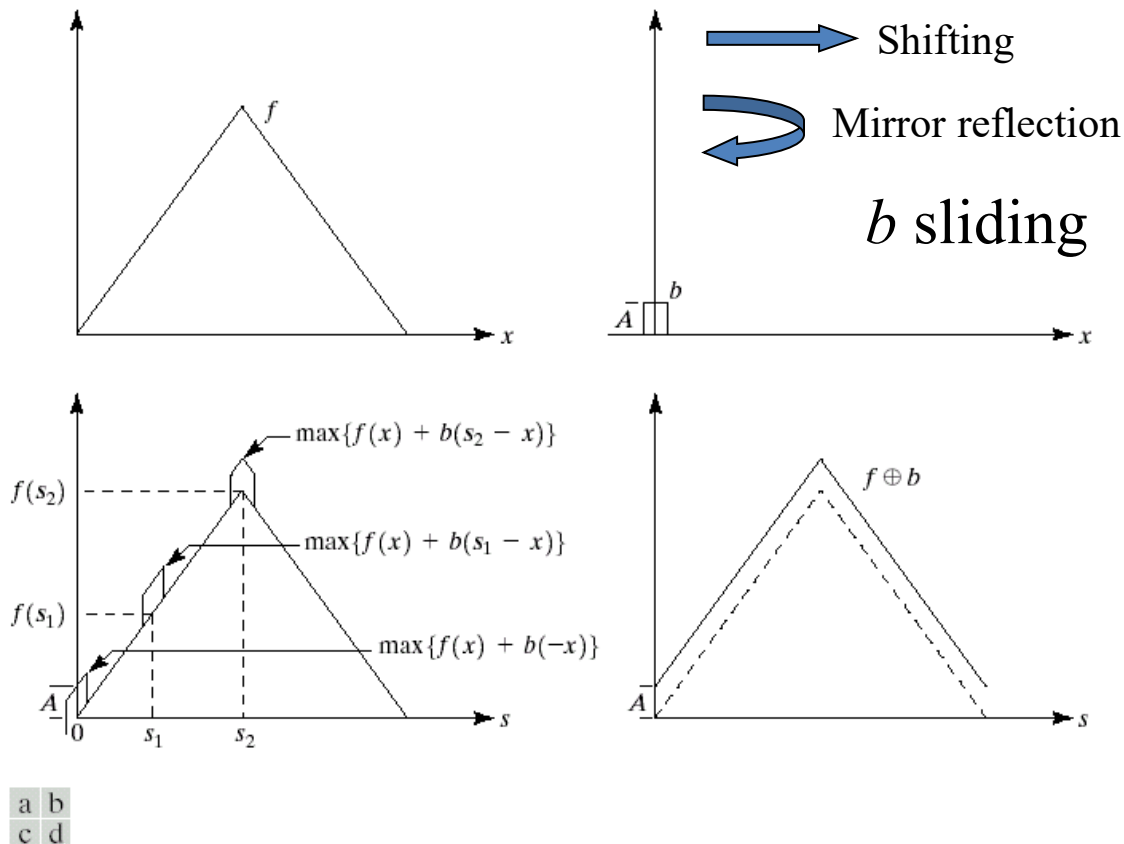
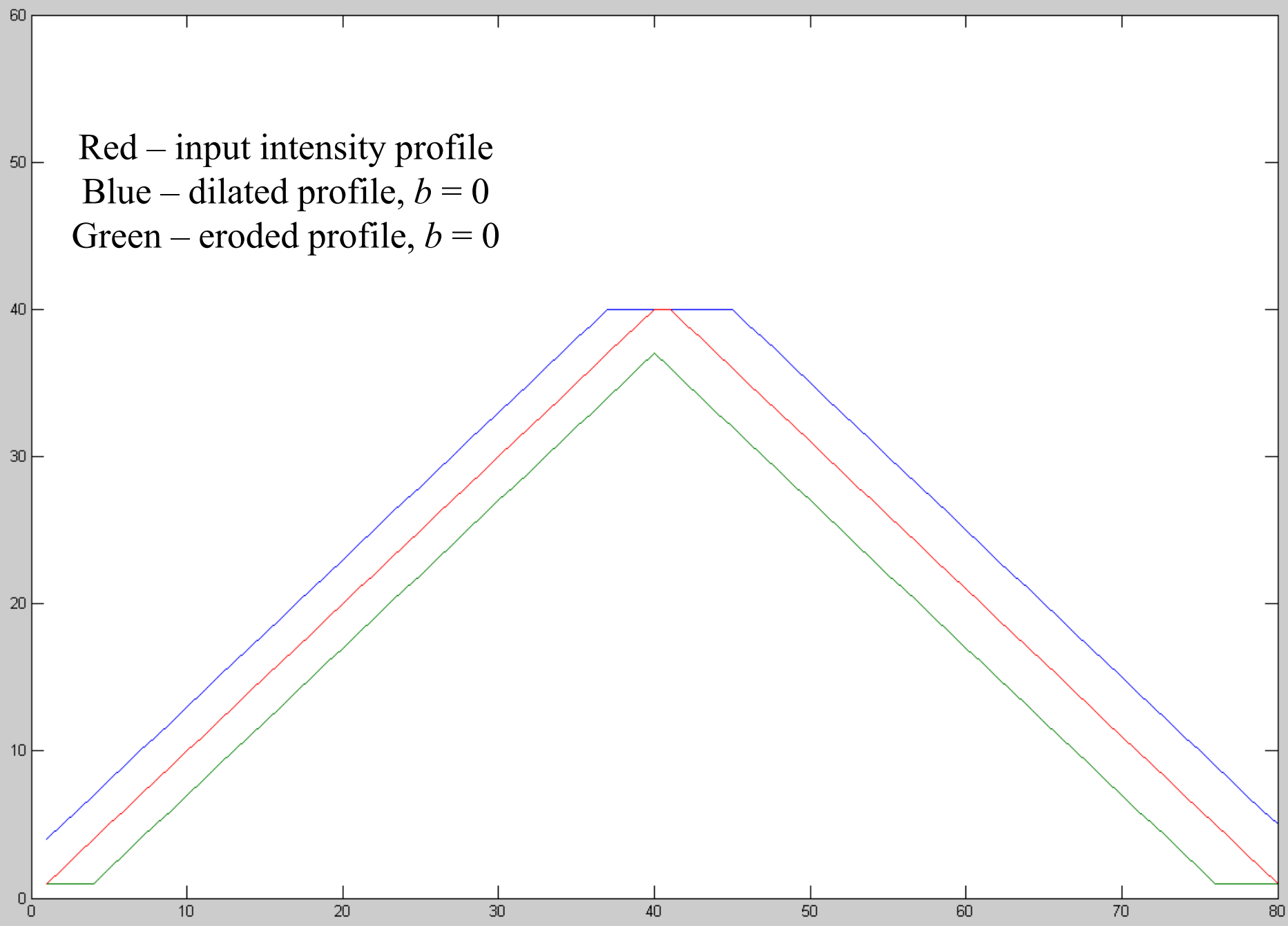


FIGURE 9.27 (a) A simple function. (b) Structuring element of height A . (c) Result of dilation for various positions of sliding b past f . (d) Complete result of dilation (shown solid).



Dilation

11. The general effect of performance:

- a. If all values of the structuring element are positive, then the output image tends to be brighter than the input.
- b. Dark small details are either reduced or eliminated.



a b

FIGURE 9.29

(a) Original image. (b) Result of dilation.

(Courtesy of
Mr. A. Morris,
Leica Cambridge,
Ltd.)

Erosion

1. 2D gray-scale erosion of f by b is defined as

$$(f \ominus b)(s, t) = \min\{f(s + x, t + y) - b(x, y) \mid (s + x), (t + y) \in D_f; (x, y) \in D_b\}$$

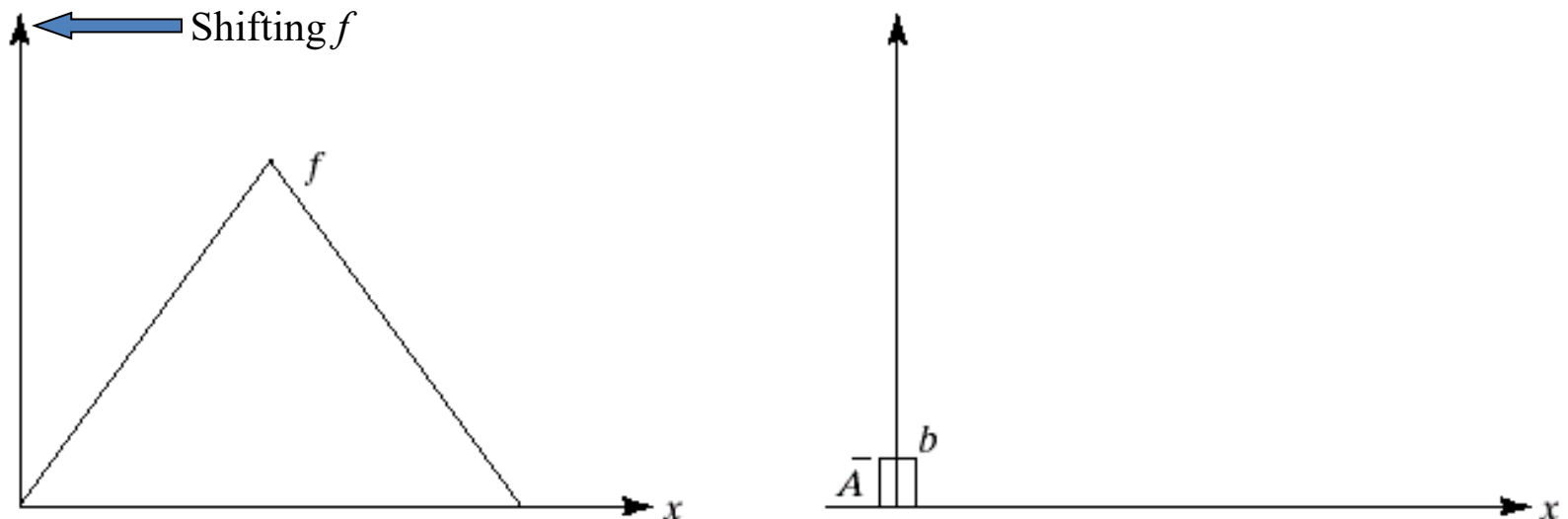
2. D_f and D_b are the domains of $f(s+x, t+y)$ and $b(x, y)$.
3. If the function b is zero, the gray-level erosion operation is the same as finding the local minimum.
4. $(s+x)$ and $(t+y)$ have to be in the domain of $f(s+x, t+y)$.
5. x and y have to be in the domain of $b(x, y)$.
6. Points 3 and 4 are similar to the conditions in the binary definition of erosion, which ensure that the structuring element is completely contained within *the set being eroded*.

Erosion

7. 1D gray-scale erosion of f by b is defined as

$$(f \ominus b)(s) = \min\{f(s+x) - b(x) \mid (s+x) \in D_f; x \in D_b\}$$

8. $f(s+x)$ is equivalent to shifting to the left (right) when the value of s is positive (negative).



Erosion

9. As f sliding by b is no different with b sliding by f , the actual mechanics of gray-scale dilation are easier to visualize if b is the function that slides past f .

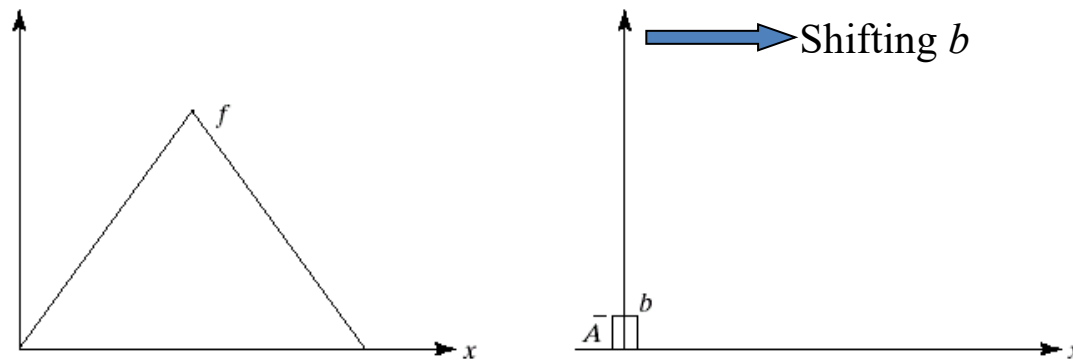
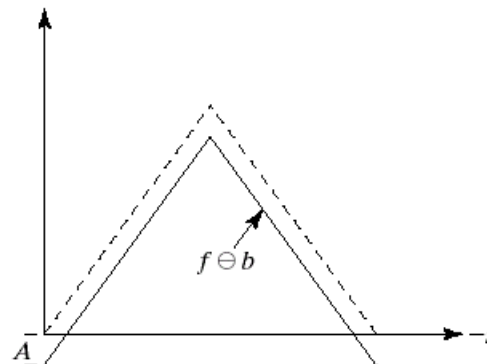


FIGURE 9.28
Erosion of the
function shown in
Fig. 9.27(a) by the
structuring
element shown in
Fig. 9.27(b).



10. The general effect of performance:

- a. If all values of the structuring element are positive, then the output image tends to be darker than the input.
- b. The effect of bright small details in the input image that are smaller in area than the structuring element is reduced.



Original image



Results of erosion

Erosion

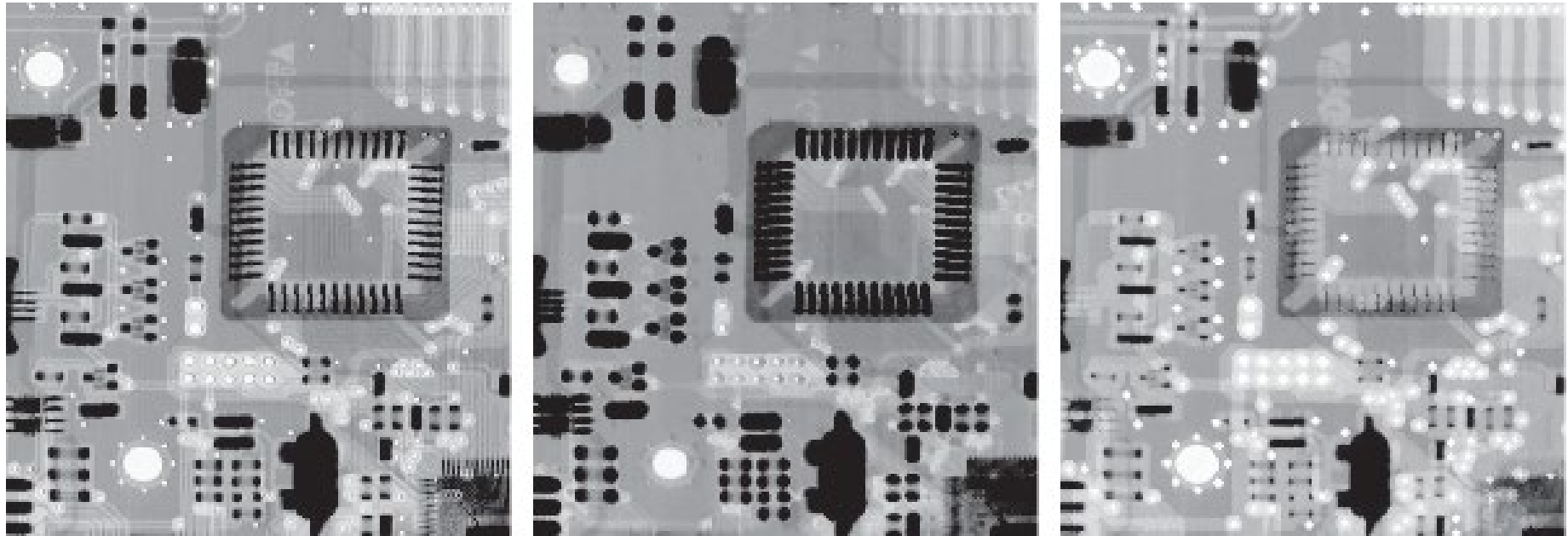
11. Gray-scale dilation and erosion are duals with respect to function complementation and reflection.

$$(f \ominus b)^c(s, t) = (f^c \oplus \hat{b})(s, t)$$

where

$$f^c = -f(x, y)$$

$$\hat{b} = b(-x, -y)$$



a b c

FIGURE 9.37

(a) Gray-scale X-ray image of size 448×425 pixels. (b) Erosion using a flat disk SE with a radius of 2 pixels. (c) Dilation using the same SE. (Original image courtesy of Lixi, Inc.)

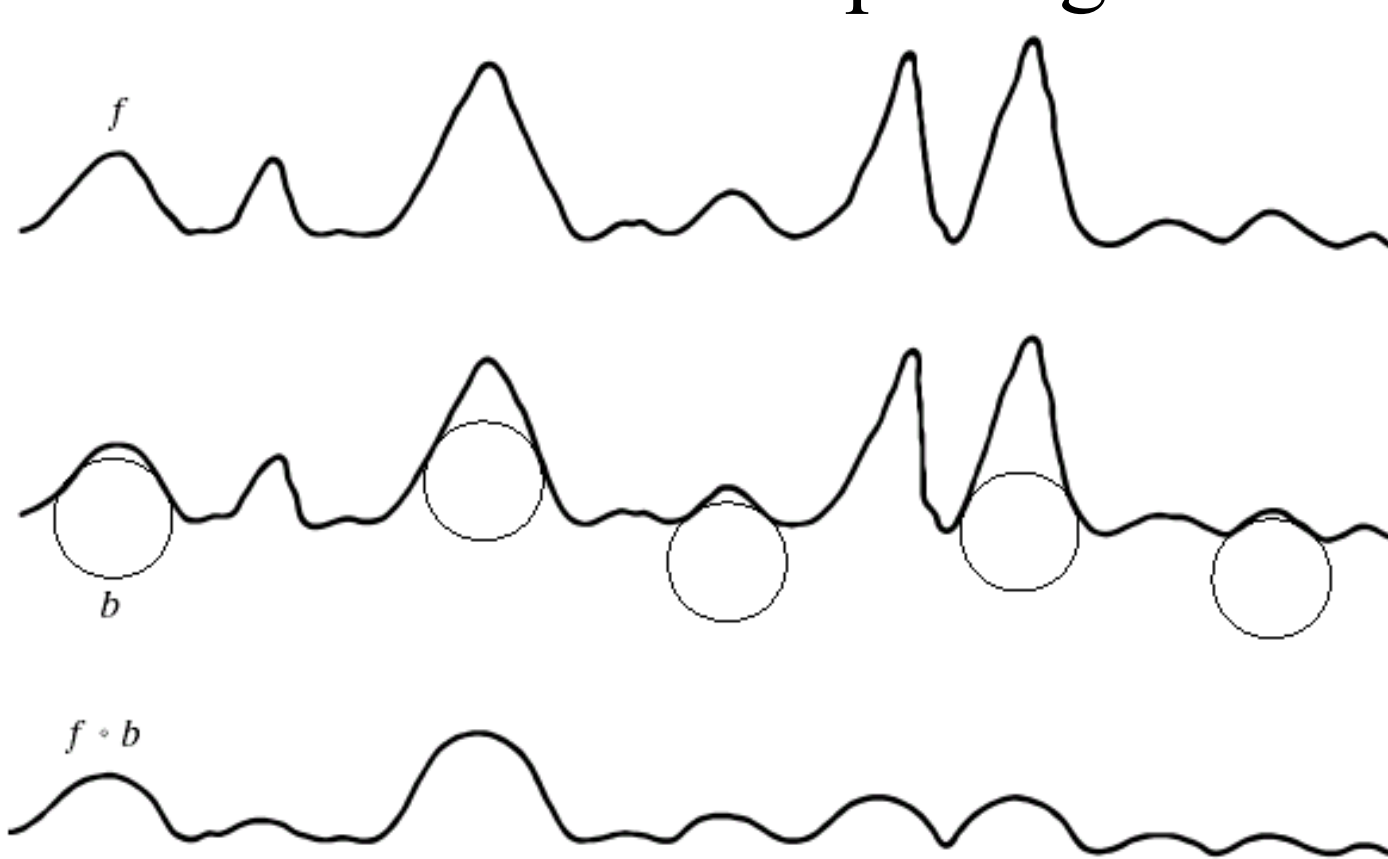
Opening

1. The opening of image f by structuring element b is given by

$$f \circ b = (f \ominus b) \oplus b$$

2. As in the binary case, opening is simply the erosion of f by b , followed by a dilation of the result by b .
3. Suppose that we open f by a spherical structuring element b , (a ‘rolling ball’).
4. It is equivalent to pushing the ball against the underside of the surface, while at the same time rolling it so that the entire underside of the surface is traversed.
5. $f \circ b$ = surface of the highest points reached by any part of the sphere as it slides over the entire undersurface of f .

Opening



a
b
c
d
e

FIGURE 9.30

(a) A gray-scale scan line.

(b) Positions of rolling ball for opening.

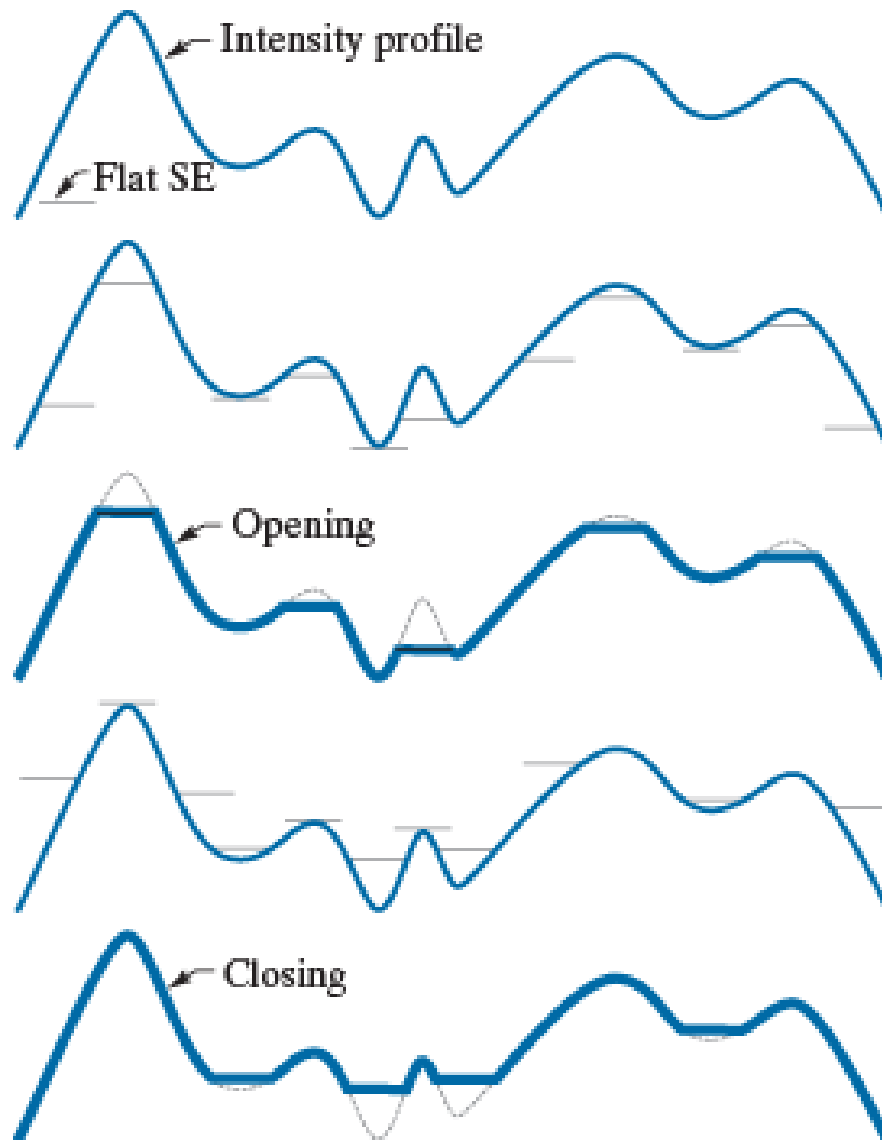
(c) Result of opening.

(d) Positions of rolling ball for closing.

(e) Result of closing.

- All the peaks that were narrow with respect to the diameter of the ball were reduced in amplitude and sharpness.
- Opening operations usually are applied to remove small light details, while leaving the overall gray levels and larger bright features relatively undisturbed.

Opening



a
b
c
d
e

FIGURE 9.38

Grayscale opening and closing in one dimension.

(a) Original 1-D signal.

(b) Flat structuring element pushed up underneath the signal.

(c) Opening.

(d) Flat structuring element pushed down along the top of the signal.

(e) Closing.

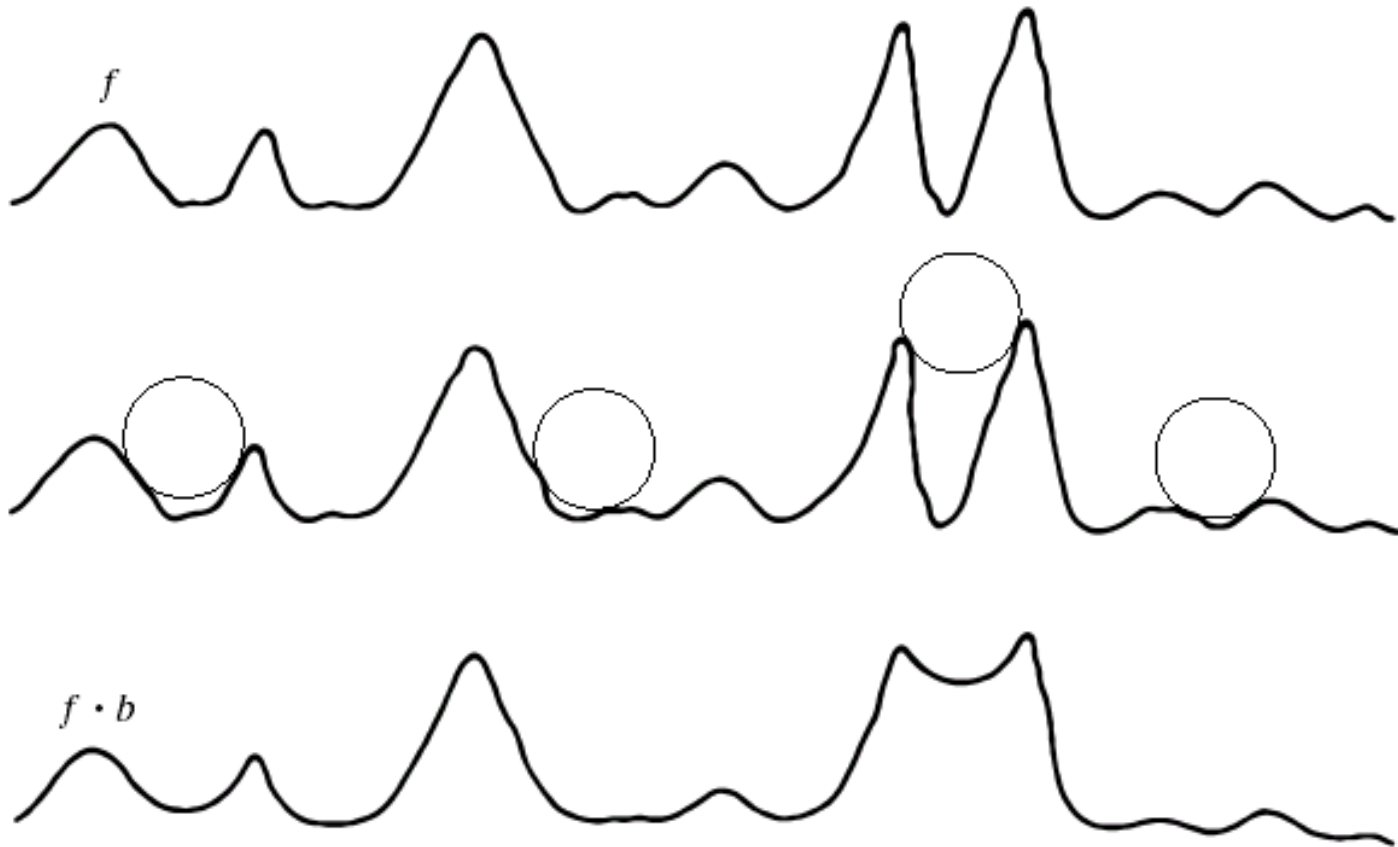
Closing

1. The closing of image f by structuring element b is given by

$$f \bullet b = (f \oplus b) \ominus b$$

2. As in the binary case, closing is simply the dilation of f by b , followed by an erosion of the result by b .
3. Suppose that we close f by a spherical structuring element b , (a ‘rolling ball’).
4. It is equivalent to rolling the ball on the top of the surface so that the entire surface is traversed.
5. $f \bullet b$ = surface of the lowest points reached by any part of the sphere as it slides over the entire surface of f .

Closing

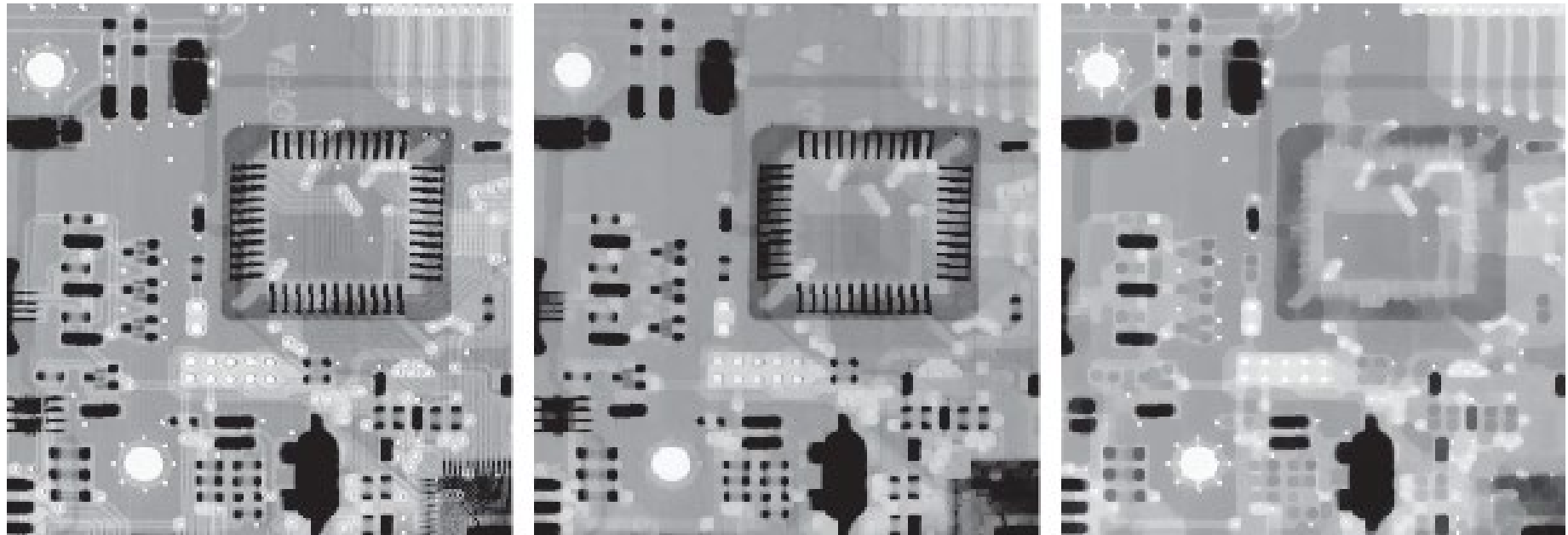


- Peaks essentially are left in their original form.
- Closing is generally used to remove dark details from an image, while leaving bright features relatively undisturbed.



a b

FIGURE 9.31 (a) Opening and (b) closing of Fig. 9.29(a). (Courtesy of Mr. A. Morris, Leica Cambridge, Ltd.)



a b c

FIGURE 9.39

(a) A grayscale X-ray image of size 448×425 pixels.

(b) Opening using a disk SE with a radius of 3 pixels.

(c) Closing using an SE of radius 5.

Duality

1. The opening and closing for gray-scale images are duals with respect to complementation and reflection.

$$(f \bullet b)^c(s, t) = (f^c \circ \hat{b})(s, t)$$

where

$$\begin{aligned} f^c &= -f(x, y) \\ -(f \bullet b) &= (-f \circ \hat{b}) \end{aligned}$$

Applications: Morphological smoothing

1. Morphological opening followed by a closing.
2. The net result is to remove or attenuate both bright and dark artifacts or noise.



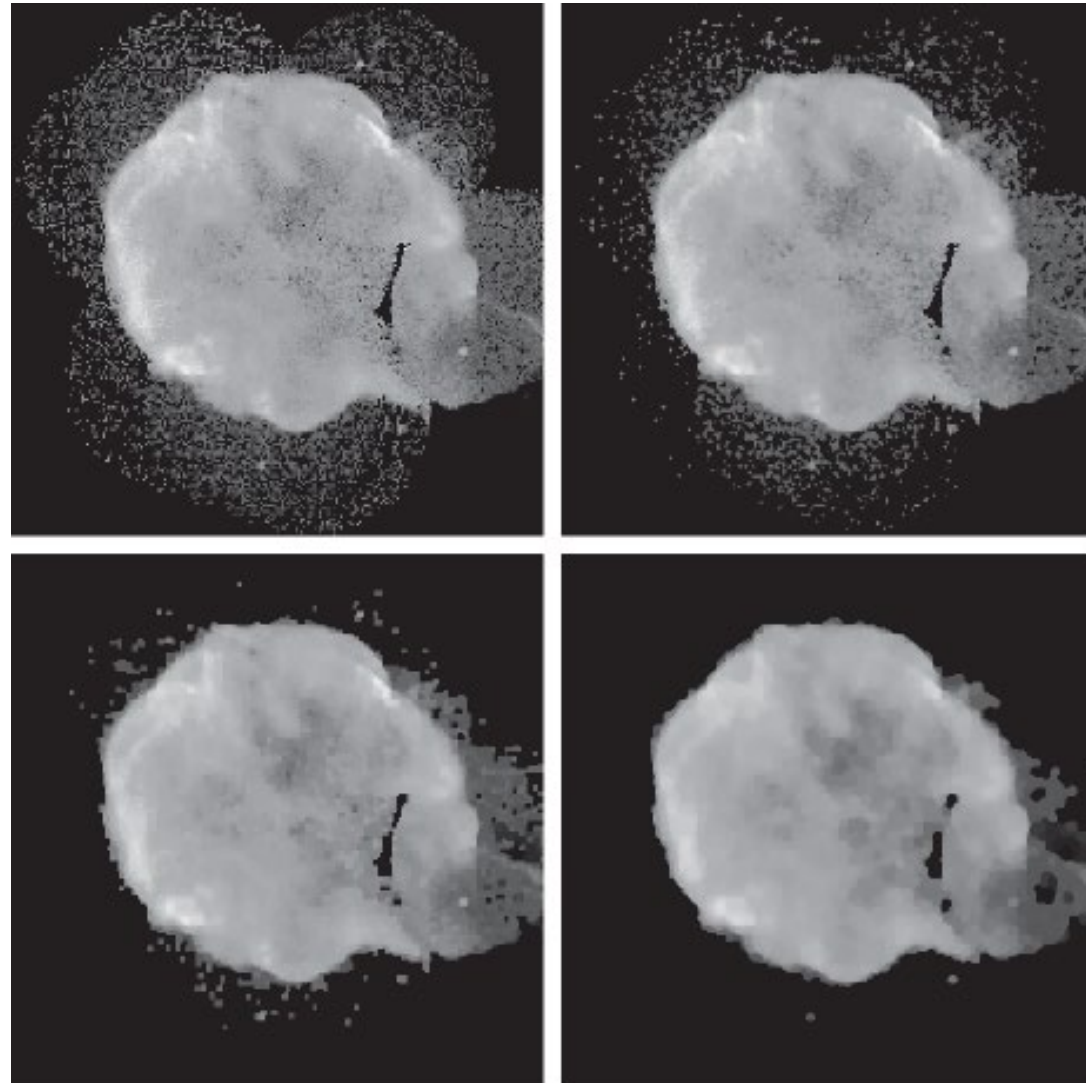
FIGURE 9.32 Morphological smoothing of the image in Fig. 9.29(a). (Courtesy of Mr. A. Morris, Leica Cambridge, Ltd.)

Applications: Morphological smoothing

a	b
c	d

FIGURE 9.40

(a) 566×566 image of the Cygnus Loop supernova, taken in the X-ray band by NASA's Hubble Telescope.
(b)–(d) Results of performing opening and closing sequences on the original image with disk structuring elements of radii, 1, 3, and 5, respectively. (Original image courtesy of NASA.)



Applications: Morphological gradient

1. Morphological gradient is defined as

$$g = (f \oplus b) - (f \ominus b)$$

2. It highlights sharp gray-level transitions in the input image.

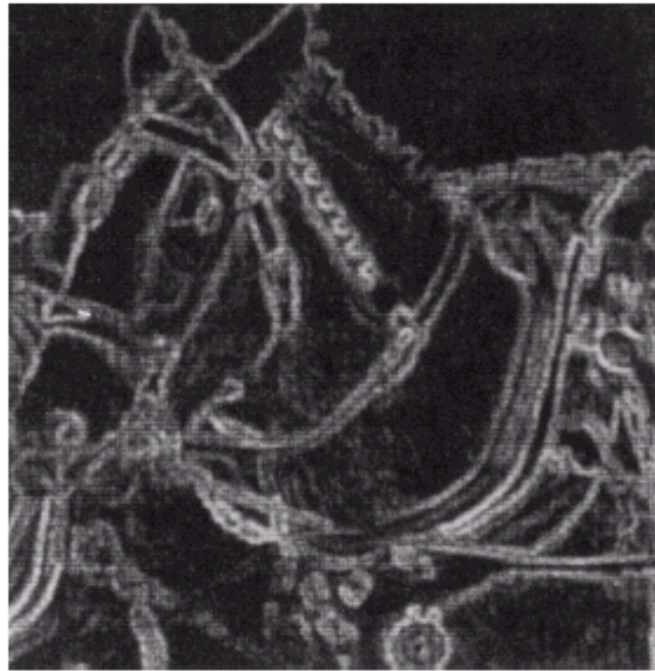


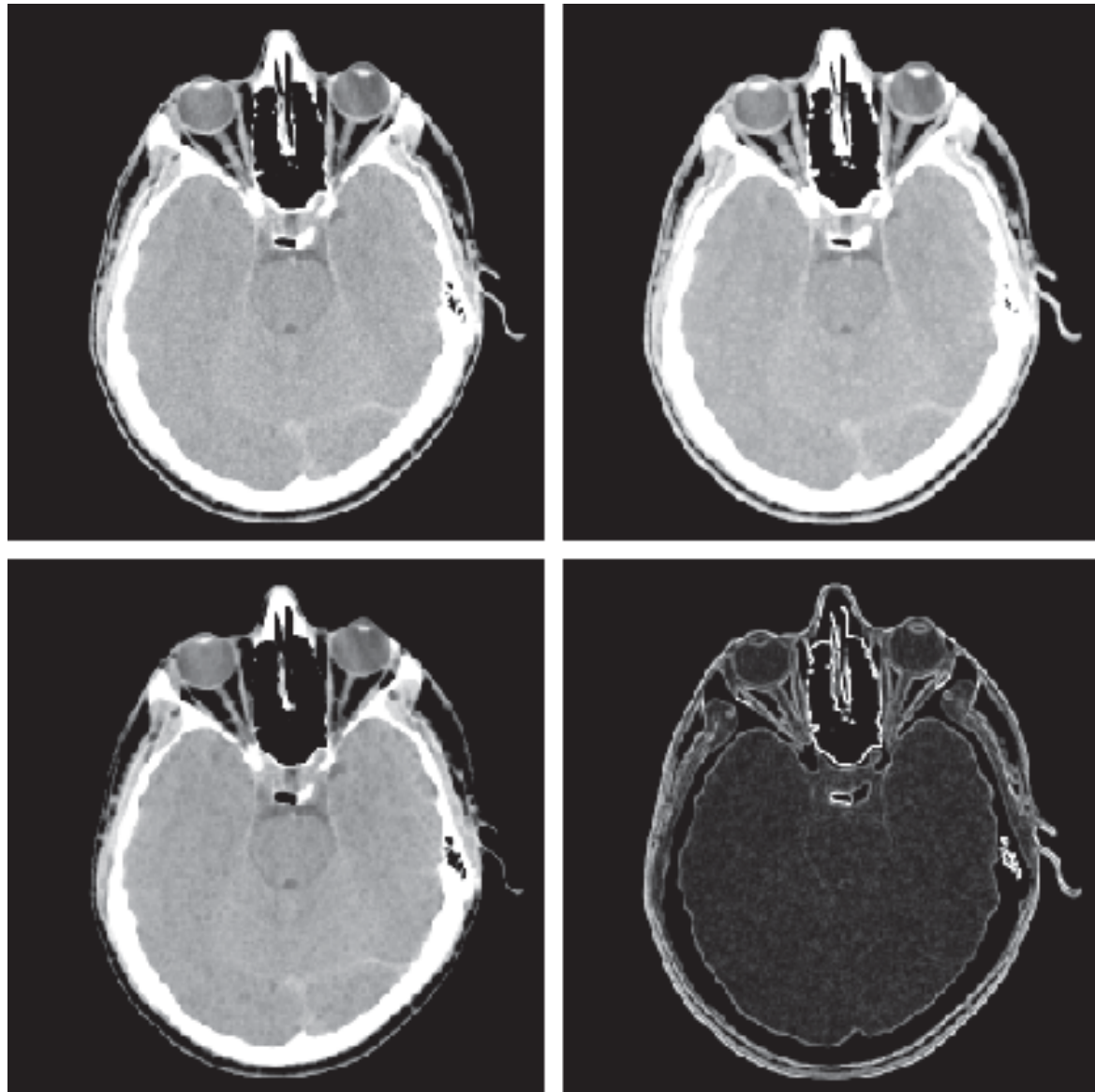
FIGURE 9.33 Morphological gradient of the image in Fig. 9.29(a). (Courtesy of Mr. A. Morris, Leica Cambridge, Ltd.)

Applications: Morphological gradient

a b
c d

FIGURE 9.41

(a) 512×512 image of a head CT scan.
(b) Dilation.
(c) Erosion.
(d) Morphological gradient, computed as the difference between (b) and (c). (Original image courtesy of Dr. David R. Pickens, Vanderbilt University.)

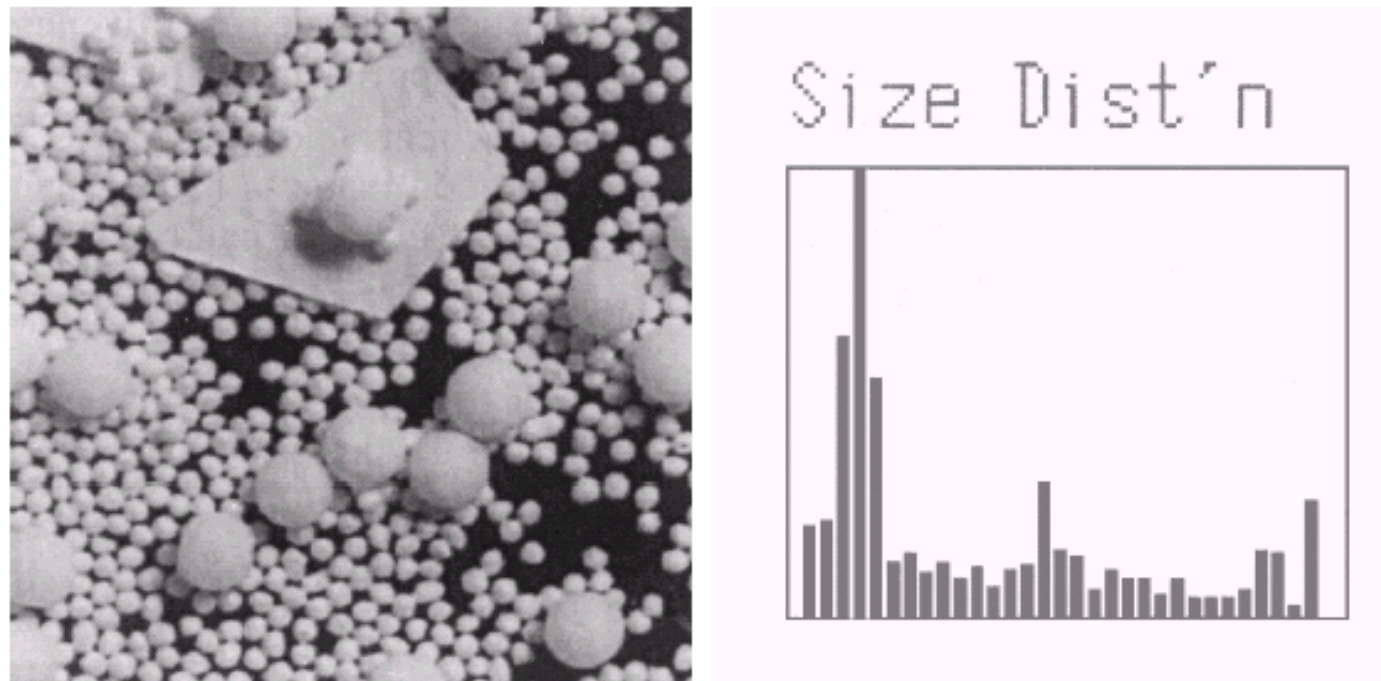


Applications: Granulometry

1. It deals with determining the size distribution of particles in an image.
2. Opening operations with structuring elements of increasing size are performed on the original image.
3. The difference between the original image and its opening is computed after each pass when a different structuring element is used.
4. At the end of the process, these differences are normalized and then used to construct a histogram of particle-size distribution.
5. It is very useful for describing regions with a predominant particle-like character.

[http://en.wikipedia.org/wiki/Granulometry \(morphology\)](http://en.wikipedia.org/wiki/Granulometry_(morphology))

Applications: Granulometry



a b

FIGURE 9.36

(a) Original image consisting of overlapping particles; (b) size distribution.

(Courtesy of Mr. A. Morris, Leica Cambridge, Ltd.)

Applications: Granulometry

a b c
d e f

FIGURE 9.43

(a) 531×675 image of wood dowels.

(b) Smoothed image.

(c)–(f) Openings of (b) with disks of radii equal to 10, 20, 25, and 30 pixels, respectively.

(Original image courtesy of Dr. Steve Eddins, MathWorks, Inc.)

